

TECHNICAL REVIEW DOCUMENT
For
RENEWAL / MODIFICATION of OPERATING PERMIT 95OPBO082

CEMEX Construction Materials South, LLC – Lyons Cement Plant
Boulder County
Source ID 0130003

Prepared by Jacqueline Joyce
March 2015 and February through June 2016
Revised August, October and December 2016
Revised February 2017 to correct permit condition number references in Appendix B

Reviewed by:

Operating Permit Supervisor:
Field Services Unit:

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I. Purpose:

This document will establish the basis for decisions made regarding the applicable requirements, emission factors, monitoring plan and compliance status of emission units covered by the renewed Operating Permit proposed for this site. The current Operating Permit was issued March 1, 2008. The expiration date for the permit was March 1, 2014. However, since a timely and complete renewal application was submitted, under Colorado Regulation No. 3, Part C, Section IV.C all of the terms and conditions of the existing permit shall not expire until the renewal Operating Permit is issued and any previously extended permit shield continues in full force and operation. The source submitted a renewal application on March 1, 2012. In addition, the source submitted applications on August 4, 2014, March 9 (revised on April 16), June 19 (revised on August 4), August 21, and October 30, 2015 to modify the permit to reflect installation of non-selective catalytic reduction on the kiln (August 4, 2014 modification), to permit the equipment to support the additional kiln control devices (March 9, 2015 modification), to increase throughput through the Dowe Flats Quarry (June 19, 2015 modification), to include the extension to the compliance date for the requirements in 40 CFR Part 63 Subpart LLL (August 21, 2015 modification) and to incorporate requirements from a federal Consent Decree (October 30, 2015 modification).

This document is designed for reference during the review of the proposed permit by the EPA, the public, and other interested parties. The conclusions made in this report are based on information provided in the renewal application submitted March 1, 2012 and the modification applications submitted on August 4, 2014 and March 9 (revised April 16), June 19 (revised August 4), August 21, and October 30, 2015, additional information submitted on July 1, 2013, September 5, 2014, July 24, 2015 and April 1, April 11, May 19 June 6, August 12, August 25, October 18 and December 7, 8, 9, 14 and 20, 2016, comments on the draft permit and technical review document submitted on September 26, 2016 via e-mail, comments on the draft permit and technical review

document received via email on November 18, 2016 during the public comment period, inspection reports and various e-mail correspondence, as well as telephone conversations with the applicant. Please note that copies of the Technical Review Document for the original permit and any Technical Review Documents associated with subsequent modifications of the original Operating Permit may be found in the Division files as well as on the Division website at www.colorado.gov/cdphe/airTitleV. This narrative is intended only as an adjunct for the reviewer and has no legal standing.

Any revisions made to the underlying construction permits associated with this facility made in conjunction with the processing of this Operating Permit application have been reviewed in accordance with the requirements of Regulation No. 3, Part B, Construction Permits, and have been found to meet all applicable substantive and procedural requirements. This Operating Permit incorporates and shall be considered to be a combined construction/operating permit for any such revision, and the permittee shall be allowed to operate under the revised conditions upon issuance of this Operating Permit without applying for a revision to this permit or for an additional or revised construction permit.

II. Description of Source

This facility manufactures portland cement. Limestone and other raw materials extracted from the Dowe Flats quarry are processed through a primary crusher at the Dowe Flats quarry. The crushed material is transported to the plant on a 2.0 mile belt conveyor system and discharged to a stockpile. The stockpiled material is placed on a belt by means of a front end loader to be processed through a primary crusher, the dryer, and a secondary crusher. The material from the secondary crusher is stored in raw material storage silos. These storage silos contain silica and iron ore and various quarried raw materials. Material from these storage silos is discharged to weigh belts for the formulation of a desired product. The weigh belts discharge to the raw mill. The raw mill mixes and crushes the blended materials and delivers the homogenized material to storage silos. The homogenized material from the storage silos is delivered to the calciner portion of the kiln. Pulverized coal from the coal mill is fired at the bottom of the flash calciner. The calcined material from the calciner then enters the rotary kiln, which is located at a slight incline along its horizontal axis. The material travels towards the clinker discharge end where additional pulverized coal is fired for the clinkering process. The clinker is discharged from the kiln into the clinker cooler where it is cooled by air forced through the clinker bed by undergrate fans. A good percentage of this air is recovered for use as primary air in the kiln combustion process. The cooled clinker is then moved to internal storage in an A-Frame building, or outside storage stockpiles. The stored clinker is the raw material for the finish mill. In the finish mill the clinker is combined with gypsum, ground to a fine material and stored in product silos. The material in the product silos can be loaded for bulk transport, or sent to a packaging system. From an over-all perspective, the manufacturing process may be viewed as two segments -- clinker production and cement production. The clinker storage allows the two processes to operate at different production rates. During periods of low demand for cement, clinker is accumulated. If cement is in high demand, the clinker

production can be supplemented by purchase of clinker from other sources. The overall result is the clinker production can operate at a rather steady rate, while the cement production can operate in response to the current or projected demands.

In late December 2014, early January 2015 the kiln was equipped with a non-selective catalytic reduction (SNCR) system to reduce NO_x emissions. The SNCR was required by a Consent Decree (09-cv-0019-MSK-MEH) filed April 19, 2013. In addition, activated carbon and lime injection systems will be installed in 2015 in order to meet future requirements under 40 CFR Part 63 Subpart LLL, National Emission Standards for Hazardous Air Pollutants for the Portland Cement Manufacturing Industry.

The facility is located near Lyons, 12 miles north of Boulder. The area in which the facility is located is classified as attainment/maintenance for particulate matter less than 10 microns (PM₁₀). Under that classification, all SIP-approved requirements for PM₁₀ will continue to apply in order to prevent backsliding under the provisions of Section 110(l) of the Federal Clean Air Act. This area is classified as nonattainment for ozone and is part of the 8-hr Ozone Control Area as defined in Colorado Regulation No. 7, Section II.A.1.

There are no affected states within 50 miles of the plant. Rocky Mountain National Park, Rawah Wilderness Area and Eagle's Nest Wilderness Area are Federal Class I designated areas within 100 kilometers of the plant.

The summary of emissions that was presented in the Technical Review Document (TRD) for the previous renewal permit issuance has been modified to more appropriately identify the **potential to emit (PTE) and actual emissions** of both criteria and hazardous air pollutants. Emissions (in tons/yr) at the facility are as follows:

Pollutant	Emissions (tons/yr)	
	Potential	Actual
PM	590.68 [451.28]	266.97
PM ₁₀	386.86 [451.28]	126.42
SO ₂	1,377.23 [132.23]	23.29
NO _x	2,667.44 [919.44]	1,049.67
CO	471.19	306.09
VOC	282.80	4.91
Lead (Pb)	6	0.93
HAPs	11.43	3.5

Values above in brackets "[]" reflect the PTE when the new PC MACT and Regional Haze (RH) requirements take effect. Note that the PC MACT total hydrocarbon (TCH) limits (kiln and dryer) and HCl limits (kiln) are not reflected as these are ppm limits and not easily reduced to a tons/yr limit.

Potential to emit is summarized by emission unit and/or emission group in the table shown on pages 58-59. The detailed table on pages 58-59 also indicates the basis for the potential emissions

Actual emissions are based on the most recent APEN on file and a breakdown of actual emissions by emission unit and/or emission group is shown in the table on pages 60-61.

National Emission Standards for Hazardous Air Pollutants (NESHAP) for Source Categories

The Lyons Cement Plant was operated as a major source of hazardous air pollutant (HAP) emissions upon the first compliance date for the Portland Cement NESHAP. In the revised Title V permit application dated October 23, 1998, the source revised the hazardous air pollutant emissions estimates. In both the original and revised Title V permit applications, the kiln was the only source presumed to emit hazardous air pollutants and hydrochloric acid (HCl) was the significant HAP. In the October 23, 1998 Title V permit application, the source used an emission factor for HCl from a September 1996 Portland Cement Association report to EPA. This emission factor (0.0381 lb/ton clinker) indicated that emissions from HCl were above 10 tons/yr thus the facility was a major source for HAPs. A performance test conducted in 2002 indicated that HCl emissions were much lower, slightly less than 4 tons/yr. However, since there is a high probability that the HCl emissions may vary and the source has never taken limits to reflect that HCl emissions may be below the major source level, the Division considers that the source is a major source for other NESHAPs issued after the Portland Cement NESHAP.

Beyond the Portland Cement NESHAP, the applicability of various NESHAPs (also referred to as “MACT requirements”) has not been discussed in detail in the previous technical review documents. Since the issuance of the March 1, 2008 renewal permit, additional MACT standards have been promulgated that may apply to this facility and are being discussed here.

Portland Cement Manufacturing Industry (40 CFR Part 63 Subpart LLL)

The Portland Cement Manufacturing NESHAP (also referred to as “PC MACT”) was promulgated on June 14, 1999 and the facility had to comply with the PC MACT beginning on June 12, 2002 (these requirements included particulate matter (PM) and opacity limitations for the various equipment). On December 15, 2000, the U. S. Court of Appeals for the D. C. Circuit, remanded parts of the 1999 requirement and the court instructed EPA to consider setting standards for HCl, mercury (Hg), total hydrocarbons (THC) and metal HAPs.

On December 8, 2006, EPA issued final amendments to the PC MACT and these revisions set emissions limits for Hg and THC from new kilns (commenced construction after December 5, 2005) and set work practice standards to address Hg and THC from existing kilns. These amendments did not set emission limits for HCl. However, on December 8, 2006 EPA announced that it would reconsider the emissions limits for Hg and THC for new kilns.

On September 10, 2010, EPA issued revisions to the PC MACT that set lower PM limits for new and existing kilns, revised the Hg and THC limits for new kilns, set limit for Hg, and THC on existing kilns and set HCl limits on new and existing kilns. Existing units would have 3 years to comply with the revised PC MACT requirements (compliance date September 9, 2013). In May 2011 EPA indicated that they would reconsider several issues raised in a petition and in December 2011, the U. S. Court of Appeals for the D. C. Circuit, remanded the rule to EPA to address the development of standards for commercial and industrial solid waste incinerators (CISWI) and the PC MACT (some kilns would fall under the CISWI category and should not be included in setting PC MACT standards) and to stay the requirements for open clinker piles that were set in the September 10, 2010 requirements.

On February 12, 2013, EPA promulgated revisions to the PC MACT, with a compliance date of September 9, 2015 for existing sources (for the February 12, 2013 standards), except that compliance with the outdoor clinker storage requirement was February 12, 2014. The applicable requirements from the February 12, 2013 requirements will be included in the permit. Note that the source is subject to the December 8, 2006 requirements until the February 12, 2013 standards take effect on September 9, 2015. Since the renewal permit will be issued after September 9, 2015, only the February 12, 2013 requirements will be included in the permit.

It should be noted that proposed revisions to the PC MACT were published in the Federal Register on November 19, 2014 to make technical corrections and to remove the rule provisions for establishing an affirmative defense. These revisions were finalized in the Federal Register on July 27, 2015.

The appropriate PC MACT requirements will be included in the permit. A more detailed discussion of the PC MACT requirements are included later in this document under the discussion of NSPS Subpart F requirements and the March 1, 2012 renewal application.

Reciprocating Internal Combustion Engines (40 CFR Part 63 Subpart ZZZZ) (40 CFR Part 63 Subpart ZZZZ)

The reciprocating internal combustion engine (RICE) MACT was published in the Federal Register on June 15, 2004 and under this rulemaking only RICE that were > 500 hp and located at major sources of HAPS were subject to the requirements. Subsequent revisions were made to the RICE MACT to address new engines $\leq$ 500 hp located at major sources and new engines of all sizes at area sources (final revisions published January 18, 2008), existing compression ignition engines $\leq$ 500 hp at major sources and all sizes at area sources (final revisions published March 3, 2010) and existing spark ignition engines $\leq$ 500 hp at major sources and all sizes at area sources (final revisions published August 20, 2010). Revisions were made on January 30, 2013 and these revisions primarily changed the requirements for engines greater than 500 hp located at area sources and the operating requirements for emergency engines.

The D. C. Circuit Court issued a mandate on May 4, 2016 for vacatur for certain requirements allowing emergency engines to operate for limited hours for demand response. So it should be noted that additional revisions to the requirements in 40 CFR Part 63 Subpart ZZZZ are expected to be made in response to issues related to legal action associated with the allowable hours of operation provisions for emergency engines regarding engines used for demand response. If such revisions are finalized prior to issuance of the permit, they will be included in the permit.

No stationary internal combustion engines are included in the insignificant activity list of the current Title V permit (last revised April 1, 2013). However, the source submitted an APENs for several engines on July 1, 2013, under the assumption that APENs needed to be filed due to changes in Regulation No. 3. Upon review of the July 1, 2013 submittal, an APEN was only required for one engine, the A-pit pump, since actual, uncontrolled emissions from this engine were above the APEN de minimis level (1 ton/yr of NO_x).

However, all of the engines are subject to the RICE MACT requirements; therefore, all of the engines need to be included in Section II of the permit. The appropriate RICE MACT requirements for the engines will be included in the permit.

Industrial, Commercial and Institutional Boilers and Process Heaters MACT (40 CFR Part 63 Subpart DDDDD)

The final rule for industrial, commercial and institutional boilers and process heaters (referred to as the Boiler MACT) was published in the Federal Register on September 13, 2004. Due to the vacatur of the 2004 rule, EPA was required to re-promulgate requirements for this source category. EPA promulgated National Emission Standards for Hazardous Air Pollutants for Major Sources: Industrial, Commercial, and Institutional Boilers and Process Heaters on March 21, 2011. These requirements apply to boilers and process heaters that are located at major sources of HAPs.

There is no de minimis level specified in the Subpart DDDDD requirements and there is fuel-burning equipment (propane portable heaters) identified in the insignificant activity list that is potentially subject to these requirements. However, these units do not meet the definition of boiler or process heater specified in the rule (the definition of process heater excludes units used for comfort or space heat). The source has indicated that there are no process heaters at this facility. Therefore, the Boiler MACT only applied to the auxiliary boiler.

New Source Performance Standards (NSPS)

EPA has promulgated NSPS requirements for new source categories since the issuance of the first renewal permit for this facility. NSPS requirements generally only apply to new or modified equipment and the Division is not aware of any modifications to existing equipment or additions of new equipment that would render equipment at this facility subject to NSPS requirements.

Because the recently promulgated NSPS requirements address equipment that may not be subject to APEN reporting or minor source construction permit requirements and, the applicability of some of the newly promulgated or proposed requirements are being addressed here.

NSPS Subpart F – Standards of Performance for Portland Cement Plants

The initial issuance of the PC MACT requirements specifically stated that any affected source subject to PC MACT requirements was exempted from any otherwise applicable requirements in NSPS Subpart F (see § 63.1356). The current Title V permit (last revised 4/1/13) includes the December 2006 PC MACT requirements, which also includes this exemption. Therefore, the current Title V permit (last revised 4/1/13) does not include any NSPS Subpart F requirements.

When the PC MACT was revised on September 9, 2010, the language in § 63.1356 was revised to state that if an affected facility subject to this subpart has a different emission limit or other requirement for the same pollutant under another regulation in Title 40 that the source must comply with the more stringent limit and is exempt from the less stringent limit. For all sources except the kiln and clinker cooler, the opacity limit in NSPS F is more stringent than the opacity limit in the PC MACT. Thus the more stringent NSPS F opacity requirement will be included in the permit for those sources that were previously subject to NSPS F requirements.

The initial Title V permit (issued February 1, 200) indicated that the sources in Section II.5 (dryer), II.11 (various clinker and cement material handling equipment) and II.13 (CKD and raw material waste dust handling) were subject to requirements in NSPS F. In addition, the rail car unloader was initially permitted in 2006 and thus is also subject to NSPS F requirements (i.e., the equipment commenced, construction, reconstruction and/or modification after August 17, 1971).

Note that although the more stringent opacity requirement in NSPS Subpart F will be included in the permit, the Division has not included the NSPS general provisions for the PC MACT/Subpart F affected sources. The NESHAP general provisions includes similar requirements therefore, those requirements will be included in the permit.

NSPS Subpart F specifies that a source other than kilns (including alkali bypass and clinker cooler) shall follow the opacity monitoring requirements in NESHAP LLL.

It should be noted that under the NSPS general provisions, the NSPS opacity requirements do not apply during periods of startup, shutdown and malfunction (see 60.11(c)). There is a similar exemption in the NESHAP general provisions (see 63.6(h)(1)) but the Table 1 for NESHAP LLL (applicability of general provisions) indicates that this paragraph does not apply and that compliance obligations are specified in NESHAP LLL.

There is no specific language in NESHAP LLL that specifies that emission limits (including opacity) apply at all times, including periods of startup, shutdown and malfunction. NESHAP LLL includes work practice requirements for kilns, clinker coolers and raw material dryers and startup and shutdown operating hours are not included in the 30 day averages for the kiln. NESHAP LLL requires that all particulate control devices be operated during startup and shutdown (63.1338(b)(9)). NESHAP LLL also requires that any exceedances of an emission limit due to a malfunction be reported and to keep records of the malfunction that caused the exceedance (63.1354(c) and 63.1355(g)).

While it might not be clear whether the opacity limits apply during startup and shutdown, there is no continuous monitoring requirement for equipment other than the kiln and clinker cooler (a COMS is an option for the raw and finish mills, in lieu of daily visible emission observations). For sources other than the kiln and clinker cooler, startup and shutdown is not expected to be long in duration and monitoring (i.e., Method 22 observations) for these sources are not likely to be conducted during startup and shutdown periods. In addition the NESHAP only includes specific startup requirements for the kiln, except that all particulate control devices shall be operated during startup and shutdown.

For these reasons, the NSPS opacity exemption during periods of startup, shutdown and malfunction is generally not necessary or relied upon. Therefore, the Division considers that the NSPS F opacity limit for sources other than the kiln or clinker cooler is still more stringent than the NESHAP limit for those same sources and the exemption from the NSPS opacity requirement has not been included in the permit.

NSPS Subpart DDDD – Emission Guidelines for Commercial and Industrial Solid Waste Incineration (CISWI) Units that Commenced Construction on or Before November 30, 1999

The guidelines for CISWI have been revised since the first renewal permit for this facility was issued on March 1, 2008 thus the applicability of these requirements is being reviewed with this second renewal.

The CISWI emission guidelines apply to units that 1) commenced construction on or before June 4, 2010, or commenced modification or reconstruction after June 4, 2010 but no later than August 7, 2013, 2) meet the definition of CISWI unit per § 60.2875 and 3) are not exempt under § 60.2555.

The CEMEX kiln meets the criteria in item 1) and is not exempt under item 3). According to § 60.2875, a CISWI unit means “any distinct operating unit of any commercial or industrial facility that combusts, or has combusted in the preceding 6 months, any solid waste as that term is defined in 40 CFR part 241. If the operating unit burns materials other than traditional fuels as defined in §241.2 that have been discarded, and you do not keep and produce records as required by §60.2740(u), the operating unit is a CISWI unit.”

Permitted fuels for the kiln include natural gas, coal, petroleum coke and tires (whole or shredded). Natural gas, coal and petroleum coke meet the definition of traditional fuels in 40 CFR Part 241 § 241.2. Scrap tires that are not discarded and are managed under the oversight of established tire collection programs, including tires removed from vehicles and off-specification tires are considered non-hazardous secondary materials that are not solid wastes when used as fuel in a combustion unit as specified in § 241.4(a)(1).

Therefore, as long as CEMEX uses tires that are managed under the oversight of an established tire collection program, the kiln is not considered a CISWI unit. The permit will include language specifying that if CEMEX uses tires they must have been managed under the oversight of an established tire collection program or CISWI requirements would apply.

The permit also allows CEMEX to use iron slag, which may be considered a solid waste. The iron slag is used as a raw material that goes through the kiln to be made into clinker. Since the kiln is a combustion unit and iron slag, which may be considered a solid waste, passes through the kiln, it might be reasonable to assume that the iron slag may trigger CISWI requirements.

Since the iron slag is used as raw material in clinker, it seems that it would be a non-hazardous secondary material used as an ingredient in a combustion unit that is not considered a solid waste when combusted (per § 241.3(b)(2)) as long as it meets the legitimacy criteria in § 241.3(d)(2). However, in the May 17, 2011 Federal Register notice denying in part and granting in part of petitions to reconsider the rule, EPA noted the following (see page 28322)

In this analysis, the EPA finds that none of the cement kilns would have been potentially CISWI due to the use of secondary material ingredients (though some kilns would potentially have been CISWI due to secondary fuels burned). This is because none of these secondary ingredient materials identified by PCA as being used in cement kilns is considered to be combusted.

The language in the preamble to the May 17, 2011 Federal Register is based on an EPA memorandum to the rulemaking docket dated April 25, 2011. This memorandum addressed whether materials are combusted in the kiln, noting that materials that are used as ingredients are generally always added to the “cold” part of the kiln and thus are not combusted. The CISWI requirements were revised in February 2013 and one of the revisions was to revise the definition of “waste burning kiln” to address where combustion occurs.

The definition of “waste-burning kiln” in § 60.2875, specifies that “[s]econdary materials used in Portland cement kilns shall not be deemed to be combusted unless they are introduced into the flame zone in the hot end of the kiln or mixed with the precalciner

fuel.” The raw material, including any iron slag, is not introduced into the flame zone in the hot end of the kiln or mixed with the precalciner fuel, so iron slag is not “combusted” and thus the iron slag does not render the kiln a CISWI unit.

NSPS Subpart JJJJ – Stationary Spark Ignition Engines

NSPS Subpart JJJJ applies to stationary spark ignition engines that commenced construction, reconstruction or modification after June 12, 2006 and were manufactured after specified dates. The date the engine commenced construction is the date the engine was ordered by the owner/operator. There are no spark ignition engines at this facility that commenced construction, reconstruction or modification after June 12, 2006, therefore, the requirements in NSPS Subpart JJJJ do not apply.

NSPS Subpart IIII – Stationary Compression Ignition Engines

NSPS Subpart IIII applies to stationary compression ignition engines that commenced construction, reconstruction or modification after July 11, 2005 and were manufactured after specified dates. The date the engine commenced construction is the date the engine was ordered by the owner/operator. In an October 18, 2016 email, the source provided information for a new compression ignition engine that was installed to provide power to the flood response pumps. This engine has been included in the permit.

New and/or Revised Colorado Regulations

The following Colorado Regulations have been revised or are in the process of being revised since the issuance of the first renewal permit (issued March 1, 2008), thus the applicability of these requirements is being addressed here

Colorado Regulation No. 7, Sections XII – Requirements for Oil and Gas Operations in the 8-hour Ozone Control Area or any Ozone Non-Attainment or Attainment/Maintenance Areas and Section XVIII - Natural Gas-Actuated Pneumatic Controllers Associated with Oil and Gas Operations (Statewide)

These requirements do not apply since oil and gas operations do not occur at this facility.

Colorado Regulation No. 7, Section XVI - Requirements for Engines in the 8-Hour Ozone Control Area and Section XVII – Statewide Requirements for Oil and Gas Operations and Natural Gas-Fired Engines

The requirements in Section XVI were adopted in March 2004 and apply to the 8-hour ozone control area. The requirements in Section XVII were first adopted in December 2006 (last revised February 2014) and apply statewide. The requirements in Section XVI apply to natural gas fired engines. The requirements in Section XVII include requirements for storage tanks, leak detection and repair programs, glycol dehydrators and well operations at oil and gas facilities and natural gas fired engines.

Requirements for storage tanks, leak detection and repair programs, well operations and glycol dehydrators at oil and gas facilities

These requirements do not apply since oil and gas operations do not occur at this facility.

Engine requirements

The engine requirements in Section XVI and XVII.E.3 apply to natural gas-fired engines greater than 500 hp and thus do not apply to any engines located at this facility. The engine requirements in Section XVIII.E.2 apply to new, modified and relocated engines at or above 100 hp. According to the July 1, 2013 APEN for the natural gas fired engine, the engine was manufactured before July 1, 2005, which is prior to the January 1, 2008 date for engines ≥ 100 hp and < 500 hp. In addition, the natural gas-fired engine at this facility is an emergency engine (used to rotate the kiln during a power failure) and as indicated on the APEN submitted on July 1, 2013 has actual, uncontrolled emissions below the APEN de minimis level. According to Regulation No. 7, Section XVII.E.1, the requirements in Section XVII.E do not apply to engines with emissions below the APEN de minimis level. Therefore, the engine requirements in Sections XVI and XVII.E do not apply to any engines at this facility.

Colorado Regulation No. 7 - Moderate 2008 Ozone State Implementation Plan Requirements

The Colorado Air Quality Control Commission (AQCC) is holding a rulemaking hearing in November 2016 to consider revisions to Colorado Regulation No. 7 to address the moderate ozone nonattainment classification for the 2008 ozone standard. The revisions to Regulation No. 7 are expected to primarily address sources that are major for ozone (NO_x or VOC emissions > 100 tons/yr) and thus would apply to CEMEX.

The draft renewal permit for CEMEX is expected to be in for public comment prior to the rulemaking hearing thus any revisions to Regulation No. 7 adopted by the AQCC will not be included in the permit.

Regional Haze (RH) Requirements

The kiln and dryer at this facility are subject to the regional haze requirements for best available retrofit technology (BART) and as such a BART analysis was conducted and a construction permit was issued to address the BART requirements. The BART requirements were included in Colorado Construction Permit 07BO0107B (issued April 28, 2009) and the emission limitations were included in Colorado Regulation No. 3, Part F in December 2008. The limitations included in the construction permit (which were also included in Reg 3, Part F) was part of the Division's regional haze state implementation plan (SIP) that was submitted to EPA Region 8 in 2009. EPA indicated that the SIP was not approvable; therefore, the Division addressed the issues raised by EPA and the regional haze requirements for BART units were included in Colorado Regulation No. 3, Part F, which was adopted by the AQCC in January 2011.

Since the BART analyses conducted in 2007-2008 were revised and replaced by the January 2011 changes to Regulation No. 3, Part F, CEMEX requested that their construction permit be canceled on April 10, 2013. The appropriate provisions from Regulation No. 3, Part F for the CEMEX dryer and kiln have been included in the draft permit.

It should be noted that as specified in Reg 3, Part F, Section VI.A.3, CEMEX must comply with the BART limits as expeditiously as practicable, but in no event later than five years after EPA approval of the Regional Haze SIP. Deadlines for compliance with the RH limits were established in the RH Compliance Schedule issued on October 1, 2013. The compliance dates for the RH NO_x and SO₂ limits are December 31, 2017 and July 1, 2017 for the kiln and dryer, respectively. The compliance dates for the RH PM limits are May 15, 2014 and April 17, 2014 for the kiln and dryer, respectively.

Consent Decree

A Consent Decree (09-cv-0019-MSK-MEH) was lodged in the federal District Court on April 19, 2013 related to NO_x emissions at CEMEX's Lyons Cement Plant. The Consent Decree (CD) requires CEMEX to install selective non-catalytic reduction (SNCR) on the kiln, install various monitoring systems, optimize SNCR and establish a NO_x limit for the kiln. In addition, the CD required CEMEX to submit a permit application to include certain requirements from the CD into a permit.

As discussed later in this document, a permit application was submitted and the appropriate requirements from the CD have been included in the renewal permit.

Compliance Assurance Monitoring (CAM) Requirements

CAM applies to any emission unit that is subject to an emission limitation, uses a control device to achieve compliance with that emission limitation and has potential pre-control emissions greater than major source levels. CAM was addressed in the technical review document to support the first renewal (issued March 1, 2008). A number of material handling equipment, including the kiln, were deemed to have uncontrolled emissions above the major source level and used control devices (baghouses) to meet PM and PM₁₀ emission limitations, thus CAM applied to these emission units and the CAM requirements and CAM plan were included in the March 1, 2008 renewal permit.

In the March 1, 2012 renewal application, CEMEX indicated that they believed the CAM determination was made in error and that CAM does not apply to any of the emission units as the baghouses are not considered control devices but are inherent process equipment.

As discussed in more detail later in this document, under "Source Requested Modifications, March 1, 2012 Renewal Application", the Division did not agree that CEMEX demonstrated that CAM does not apply. However, the Division did agree that

CAM does not apply to the high efficiency separator baghouse (P012, S065) and the Clinker Cooler (P008, S017). Also as indicated in the technical review document for the April 1, 2013 modification (see page 12), S022 (kiln return dust silo) is subject to CAM.

The analysis to determine potential pre-control emissions from the various sources and the CAM applicability was revised in order to properly describe the methodology and assumptions used in the analysis and to reflect the lower grain-loading “emission factors” for the emission units in Section II.11. The revised analysis begins on page 62 of this document. Note that based on this analysis, the following source is no longer subject to CAM: S032 (bottom of A-frame transfer) and the following sources are now subject to CAM: S010 (raw material grinding), S011 (raw material separator), S012 (raw mill feeders), S045 (cement finish silo A2) and S066 (cement silo A5).

Changes to the CAM requirements were made and are discussed later in this document.

Following the issuance of the first renewal permit, CEMEX was required by a Consent Decree to install SNCR on the kiln and establish a NO_x limit. The source is also required to meet a RH limit for NO_x for the kiln which relies on SNCR. The requirements to install SNCR and to address the RH NO_x limit and the NO_x limit established for the Consent Decree are being addressed in this renewal permit. The current permit requires that the kiln be equipped with a NO_x CEMS and that the CEMS be used to assess compliance with the annual NO_x emission limitation. The Division considers that since the current Title V permit requires the source to use the NO_x CEMS for compliance with emission limitations (annual NO_x limit), that CAM does not apply with respect to the RH and Consent Decree NO_x limits in accordance with the provisions in 40 CFR Part 64 § 64.2(b)(1)(iv) (the Title V permit specifies a continuous compliance method).

The renewal permit will also address the RH SO₂ limit. This limit was not based on the source installing any control technology for SO₂. However, in order to meet the PC MACT HCl limits, the source is installing a dry scrubber on the kiln. While the dry scrubber may not be necessary to meet the RH SO₂ limit, the Division would consider that CAM would not apply to the RH SO₂ limit because of the SO₂ CEMS for the same reasons as discussed above for the RH and Consent Decree NO_x limits.

Greenhouse Gases

Greenhouse gas (GHG) emissions from the Lyons Cement Plant exceed 100,000 tpy CO₂e. Future modifications greater than 75,000 tons per year CO₂e may be subject to regulation (Regulation No. 3, Part A, I.B.44).

Repealed APEN Exemptions

Since the first Title V renewal permit was processed (issued March 1, 2008) the APEN exemptions for engines – limited size and hours (Reg 3, Part A, Section II.D.1.sss) and

emergency generators – limited size and hours (Reg 3, Part A, Section II.D.1.ttt) was repealed. There are no engines included in the insignificant activity list in the current permit (last revised April 1, 2013).

However, the source submitted APENs on July 1, 2013 for a number of engines at the facility. Only one engine, the A-pit pump had emissions above the APEN de minimis level (1 ton/yr NO_x). APENs were not necessary for the remainder of the engines and thus those APENs were not logged in or processed.

III. Discussion of Modifications Made

The following discussion related to modifications is with respect to the current active permit (last revised April 1, 2013) and unless specifically noted as “new”, the condition numbers identified in this document reflect the condition numbers in the current permit (last revised April 1, 2013). Because some permit conditions in the current Title V permit have been removed, reorganized and/or reformatted as part of this permitting process, the condition numbers discussed in this document may not reflect the condition numbers in the draft Title V permit.

Source Requested Modifications

March 1, 2012 Renewal Application

The March 1, 2012 renewal application is essentially a red-lined version of the Title V permit issued at the time the application was submitted. There are some explanatory comments included in the application. The renewal application included information for a series of modifications that were processed and issued on April 1, 2013 so many of the requests included in the application were addressed with the processing of the April 1, 2013 revised permit. Since many of the changes have been included in the April 1, 2013 revised permit (or alternatively were determined to be inappropriate changes), it is somewhat more difficult to determine which changes remain to be addressed for the renewal. The Division considers that the following items from the March 1, 2012 renewal application remain to be considered for the renewal and have been addressed as follows:

Section I, Condition 1.3 and various locations in Section II - Compliance Order on Consent Requirements

CEMEX made various comments throughout the permit regarding requirements that originate from a compliance order on consent and asked to discuss such requirement(s). Section I, Condition 1.3 of the Title V permit indicates that requirements from a compliance order on consent (COC) signed February 19, 2004 (No. 2002-124) have been incorporated into the permit as new applicable requirements and survive reissuance. Thus, even though the COC has been closed the requirements from the COC that were incorporated into the permit live on. CEMEX requested revisions to the following COC requirements:

- Condition 1.3.16 was revised as requested to change the location of the camera.

The source also requested a discussion with the Division with respect to various control measures but did not propose any changes to the requirements. The Division informed the source that they must identify and justify any proposed changes to fugitive emission control measures but the source did not do so, therefore no changes were made to the remaining control measures based on the renewal application.

Section I, Condition 6, Section II.24 and Appendix H – Compliance Assurance Monitoring (CAM) Requirements

In their renewal application, the source indicated that they did not believe that CAM applied to any of the emissions units at their facility. The renewal application included three arguments to indicate that CAM does not apply: 1) the baghouses are inherent process equipment not controls, 2) the emission units are exempt from CAM because they are subject to NESHAP requirements per 64.2(b)(1)(i), or 3) the emission units are exempt from CAM because the Title V permit includes a continuous compliance determination method per 64.2(b)(1)(vi).

The Division considers items 2) and 3) are not valid reasons to consider that CAM does not apply because the noted exemptions apply with respect to emission limitations or standards. As noted during the first renewal, under item 2, the emission units are exempt from CAM for the NESHAP requirements. CAM has been included in the permit with respect to the construction permit PM and PM₁₀ limits (tons/yr limits). In addition, the Division does not consider that item 3 is appropriate because CEMEX did not present any information indicating that the Title V permit included a continuous compliance method for tons/yr PM and PM₁₀ limits.

The argument that the baghouses are inherent control equipment is potentially a valid argument. The preamble to the CAM rule spells out three criteria to look at to determine if equipment is inherent process equipment vs. a control device (October 22, 1997 Federal Register, page 5493) and one of those criteria is to compare the cost savings from the product recovery to the cost of the equipment. In the renewal application, CEMEX estimated the cost based on allowable emissions and assuming 99.9% control efficiency. The Division considers that this cost analysis isn't appropriate as it should be based on the actual quantity of material recovered, not an "assumed" potential. If the material is worth recovering, then presumably the amount of material recovered would be tracked. In addition, the cost analysis conducted by CEMEX did not compare the costs of material recovery with the cost of the equipment. Thus the analysis is not appropriate and CEMEX has not demonstrated that the baghouses are inherent process equipment and the CAM requirements will remain in the permit.

Although the Division does not agree that the majority of the baghouses are inherent process equipment, the Division does agree that the high efficiency separator baghouse (P012, S065) is inherent process equipment. The construction permit (98BO0259) indicates that the baghouse for the high efficiency separator (147,060 acfm baghouse)

is process equipment and not a control device, so that baghouse will be removed from the list of CAM sources.

In addition, during the processing of the April 1, 2013 revised Title V permit, it became apparent that while the clinker cooler is included on the construction permit for the kiln (12BO444-2) the clinker cooler is not subject to any annual PM emission limitations (just the short term NESHAP PM limit which is not subject to CAM). Therefore the clinker cooler (P008, S017) is not subject to CAM and will be removed from the list of sources subject to CAM.

Also as discussed in the technical review document for the April 1, 2013 revised Title V permit (see page 12), S022 (kiln return dust silo) is subject to CAM and will be included as CAM.

The source suggested some changes to the CAM language in the event that the Division did not agree that CAM did not apply. The source suggested that the length of the visible emissions observations be reduced from 6 minutes to 1 minute and to stipulate that an excursion for the pressure differential reading is when they are at or below 0 or above 7 inches of water for greater than 15 minutes.

The source provided further information related to the request to revise the pressure differential requirements (adding "greater than 15 minutes") in a May 19, 2016 response to a request for information and stated that this request was meant for the kiln and clinker cooler baghouses. Since the clinker cooler is not subject to CAM and the Division will rely on the PC MACT monitoring for the kiln, this requested change is no longer relevant and will not be made.

In addition, the Division considers that the request to reduce the visible emission observation to 1 minute is not appropriate and the change will not be made.

While the Division did not agree that CAM doesn't apply, the CAM requirements were revised as follows:

Section I.6 Changes

- Removed the high efficiency separator (S065, P012) and the clinker cooler (S018, P008) from the list of CAM sources in this condition.
- In addition, the following source was removed from this condition: S032 (bottom of A-frame transfer) and the following sources were added to this condition: S010 (raw material grinding), S011 (raw material separator), S012 (raw mill feeders), S045 (cement finish silo A2) and S066 (cement silo A5)

As discussed previously, the Division re-did the CAM analysis in order to document the methodology, which resulted in some sources no longer subject to CAM and others now subject to CAM.

Based on comments received on the draft permit on September 26, 2016, the south (S049) and north (S050) packers are inoperable and were removed from the permit, so they are no longer subject to CAM.

Section II.24 Changes

- CAM for the kiln will be based on the NESHAP requirements. The NESHAP requires the use of a PM CEMS which will meet the frequency of monitoring for CAM (continuous) and the NESHAP PM limit is lower than the basis for the annual (ton/yr) emission limit (the annual limit is based on 0.275 lb/ton of dry feed), thus relying on the PC MACT for CAM is appropriate.
- Language was added to the daily visible emission observations to indicate that they are not required if the equipment is not operating for more than four (4) consecutive daylight hours, as long as a pressure differential reading is taken for that day.
- Frequency of recording the pressure differential was revised to weekly (calendar week).

Appendix H Changes

- The table with the description of baghouses has been removed, as this is not considered to be necessary.
- Since CAM for the kiln has been revised to be based on the NESHAP requirements and they are specifically included in the permit, the kiln is not addressed in the CAM plan.
- The visible emission observations for the finish mill, raw material grinding and raw material separator rely on the PC MACT daily observations.
- The frequency for the pressure differential reading was revised to weekly.
- The following language was added to the visible emission observations:
 - Daily readings are not required if the equipment is not operating for more than four (4) consecutive daylight hours, as long as a pressure differential reading is taken for that day.
 - Observations may be made from one point for baghouses with stacks in close proximity to each other.

Section II.16 – Opacity – Daily Plant Walk Through

In the renewal application, the source noted that they could not find this requirement in any regulation and asked that it be removed. The daily plant walk through for opacity requirement was included as periodic monitoring, which is required for Title V permits (see Colorado Regulation No. 3, Part C, Section V.C.5.b). All the stacks are subject to

opacity requirements, thus in lieu of requiring daily visible emission checks on all stacks the Division included the requirement to do observations from different locations such that all stacks can be seen. The Division considers that this requirement is still necessary for purposes of periodic monitoring and this requirement will remain in the permit. However, the following changes have been made:

- The requirements in this condition were reformatted and reorganized to make the requirements more clear. In addition some minor language changes were made or additional language was added to clarify the requirements.
- Language was added to specify that the daily observations are not required when the plant or remote operations are not operating for more than four (4) consecutive daylight hours.

Section II.23 – Portland Cement Manufacturing NESHAP

In the renewal application, CEMEX included a “new” Condition 7 in Section I of the permit to address the revised NESHAP requirements and revised Section II.23 (Portland Cement NESHAP) to indicate that the “new” NESHAP requirements were in Section I, Condition 7. This “new” Section I, Condition 7 did not include any substantive requirements most likely since the requirements were under review at the time and the “previous” NESHAP requirements applied until the compliance date for the “new” NESHAP requirements.

Upon the request of the Division, CEMEX did submit information on May 19, 2016 indicating the emission limits and monitoring options in the Portland Cement NESHAP that they would follow and those requirements have been included in Section II.23 of the permit. Specifically for the kiln, the source will install a lime injection system and compliance with the HCl limit will be monitored using the SO₂ CEMS and compliance with the mercury limit will be monitored using a sorbent trap system. For the kiln the total hydrocarbons (THC) will be monitored to meet the THC limit and for the dryer the source will comply with the organic HAP limit and will establish a THC operating limit.

Explanation regarding the inclusion and exclusion of requirements that would reasonably be considered to apply (or not apply) is as follows:

- The current Title V permit (last revised April 1, 2013) includes the December, 2006 revised NESHAP requirements, which specified in § 63.1356 that any source subject to the provisions in NESHAP LLL was not subject to the NSPS requirements in Subpart F. When NESHAP LLL was revised on September 10, 2010, the language in § 63.1356 was revised to specify that sources subject to multiple standards in Title 40 of the CFR for the same emission unit only had to comply with the most stringent requirement.

To that end, the opacity requirement in NSPS F for all affected sources except the kiln and clinker cooler is more stringent than the opacity requirement in NESHAP LLL. The opacity standard in NSPS F requires that opacity be less than 10% and

NESHAP LLL specifies that opacity shall not exceed 10%. A method 9 opacity reading of 10% would be a violation under NSPS F but would not be a violation under NESHAP LLL.

Therefore, the NSPS F opacity limit will be included in the permit for all equipment except the kiln and clinker cooler that was previously subject to NSPS F. The initial Title V permit (issued February 1, 200) indicated that the sources in Section II.5 (dryer), II.11 (various clinker and cement material handling equipment) and II.13 (CKD and raw material waste dust handling) were subject to requirements in NSPS F. In addition, the rail car unloader was initially permitted in 2006 and thus is also subject to NSPS F requirements (i.e., construction, reconstruction and/or modification occurred after August 17, 1971).

The equipment that is subject to the more stringent NSPS F opacity requirements is still subject to certain requirements in NESHAP LLL, such as the organic HAP and work practice requirements for the dryer, operation and maintenance plan requirements and the outdoor clinker storage requirements and any related testing, monitoring, recordkeeping and reporting associated with those requirements.

63.1343 – standards for kilns, clinker coolers, raw materials dryers & open clinker storage piles

- Since the permit will not be issued prior to September 9, 2015, the requirements that were in effect prior to September 9, 2010 for existing sources (§63.1343(d)) will not be included in the permit.

63.1346 – operating limits for kilns

- Some portions of a given paragraph in Subpart LLL were removed as they were not applicable. Those changes are indicated in the citation of the specific paragraph included in the permit.
- Excluded 63.1346(a)(1) through (3) (D/F temp limit), since there is no in-line kiln/raw mill.
- 63.1346(c), (d) & (e) were not included since the kiln does not utilize either sorbent or carbon injection to meet the D/F limits. Although sorbent injection is utilized, it is used to meet the Hg emission limits.
- 63.1346(f) (use of fly ash with Hg as raw material or fuel) was not included because by the time the permit is issued, the source will have demonstrated compliance with the Hg limit. 63.1346(f) indicates that once the kiln is in compliance with a Hg limit, 63.1346(f) no longer applies.

63.1348 – compliance requirements

- The opacity standards and the D/F standard for the kiln are not new limits. These limits were in effect prior to the September 9, 2010 revisions for which the

compliance date is September 9, 2015. Since the initial performance test was conducted for D/F the initial performance test requirements in 63.1348(a)(3) were not included in the permit. Although the opacity limits are not new and the initial performance test requirements have been completed for existing equipment the requirements in 63.1348(a)(2) were included in the event that new equipment is added that is subject to the opacity requirements in Subpart LLL.

- The requirements in § 63.1348(a)(4)(iii) through (iv) (THC compliance for inline kiln/raw mill) were not included since there is no in-line kiln/raw mill at this facility.
- The requirements in § 63.1348(a)(6)(ii) (affected sources without scrubbers) were not included. The kiln at this facility is equipped with a dry scrubber and so the requirements in § 63.1348(a)(6)(i) apply.
- The requirements in 63.1348(b)(3)(i) and (ii) (COMS and bag leak detectors) were not included since the finish and raw mills are not equipped with either. Daily Method 9s will be required for these sources.
- The requirements in 63.1348(b)(5) (activated carbon injection) were not included since activated carbon injection is not used to comply with the D/F emission limits.
- The requirements in 63.1348(b)(8)(i) (HCl compliance – sources w/out scrubbers) and (ii) (HCl compliance – sources w/ scrubbers) were not included because the kiln is equipped with a scrubber and an SO₂ CEMS so the source will rely on the SO₂ CEMS as an alternative.

63.1349 – performance testing requirements

- The requirements in 63.1349(b)(1)(ix) (PM testing for kiln with in-line raw mill) was not included since the kiln is not equipped with an in-line raw mill.
- The requirements in 63.1349(b)(3)(v) and (vi) (sorbent injection requirements) were not included since sorbent injection is not used to comply with the D/F emission limits.
- The requirements in 63.1349(b)(6)(i) (sources w/scrubbers) and (ii) (sources w/out scrubbers) since the source will monitor SO₂ emissions as an alternative.
- The requirements in 63.1349(b)(7)(iii) (provisions for weighted average THC emissions for kilns with in-line raw mills) and (x) (performance test for kilns with in-line raw mills) were not included since the kiln is not equipped with an in-line raw mill.
- The requirements in 63.1349(b)(8)(iii) (provisions for weighted average HCl emissions for kilns with in-line raw mills) and (vi) (performance test for kilns with in-line raw mills) were not included since the kiln is not equipped with an in-line raw mill.

63.1350 – monitoring requirements

- The requirements in 63.1350(f)(4) (COMS or bag leak detectors on finish or raw mills) were not included since neither COMS or bag leak detectors are installed on the finish or raw mills.
- The requirements in 63.1350(g)(5) (D/F monitoring - status of raw mill for in-line raw mill/kiln) were not included since the kiln is not equipped with an in-line raw mill.
- The requirements in 63.1350(h) (sorbent injection) were not included since sorbent injection is not used to meet the D/F emission limit.
- The requirements in 63.1350(k)(1) through (3) (Hg CEMS requirements) were not included since the source will be using a Hg sorbent trap system and not a Hg CEMS.
- The requirements in 63.1350(l)(1) and (2) (HCl CEMS and parametric monitoring for wet or dry scrubber) were not included in the permit since the source will be relying on the SO₂ CEMS.
- The requirements in 63.1350(m)(5) through (11) do not apply because the source is using an SO₂ CEMS ((m)(5), (7) & (9)), does not use activated carbon for D/F limit ((m)(6)), does not use bag leak detection systems ((m)(10) and (11)) and (m)(8) is reserved.

63.1355 – recordkeeping requirements

- The electronic version of the CFR does not include a 63.1355(e) and 63.1355(e) is not marked as “reserved” (like 63.1355(d)). The September 9, 2010 version of the Federal Register, which published final revisions to Subpart LLL that include the Hg and HCl limits, includes a 63.1355(e). Final revisions to Subpart LLL were published in the Federal Register on February 12, 2013 and July 27, 2015 and neither of those revisions revised or removed 63.1355(e). The online version of the July 1, 2015 CFR (pdf version) includes 63.1355(e). Therefore, the Division assumes that the electronic CFR is in error and has included 63.1355(e) in the permit as it is shown in the September 9, 2010 Federal Register and the July 1, 2015 CFR. The July 25, 2016 revisions to Subpart LLL corrected the error and restored 63.1355(e) to the regulation.

Section II.10- Kiln

In the renewal application, the source suggested language for a 93-day wood-burning test. The suggested language addresses submittal of a notification and specifies that the Division would act on the notification within 60 days. The notification requirements are essentially the information that would be included in a permit application. The renewal application was submitted in March 2012 and since then the source has indicated no interest in pursuing a wood burning test on the kiln. In addition, the

suggested permit language includes the information that would be necessary for a permit application. Since the suggested permit language specifies a notification that would be similar to what would be required in a permit application and the source has not indicated interest in a wood-burning test, the suggested language has not been included in the permit.

The following changes were made to Section II.10 of the permit based on the renewal application:

- Minor language changes were made to various conditions to clarify or be consistent with terms.
- The source requested that the data replacement requirements in Condition 10.7 be revised for situations when data is not available due to quality assurance or quality control (QA/QC) reasons. However, in their comments on the draft permit received on September 26, 2016, the source, requested additional changes to the data replacement requirements to specifically address the flow monitors. The changes made in the draft permit, reflected in part the September 26, 2016 comments. The source submitted comments on November 21, 2016, during the public comment, and objected to the data replacement language in the draft permit. For the reasons discussed below, the Division removed the data replacement requirements.

The data replacement requirements for the kiln CEMS was not included in the underlying construction permit (12BO444-2) but was included in the initial Title V permit (issued February 1, 2000). At the time the initial Title V permit was processed, data replacement requirements were being included in construction permits that for sources that required to use CEMS.

The Division's practices regarding data replacement requirements have changed since CEMEX's initial Title V permit was issued. Data replacement requirements, if warranted, apply to annual emission limitations, not short term emission limitations. Data replacement requirements are typically being applied to sources that are electric utility steam generating units, as these units are required under the Acid Rain program to replace data when the CEMS is not providing valid data. The Acid Rain data replacement requirements are specific and thus an appropriate mechanism to use for sources subject to those requirements. For sources that are not subject to the Acid Rain requirements, it is difficult to set such specific procedures for filling in data when the CEMS does not produce valid data. Thus for sources that are not subject to the Acid Rain requirements, the Division typically does not require these sources to replace data.

In addition, where a tight synthetic minor limit has been set for a source (i.e., the source took limits to avoid PSD review), the Division may endeavor to require that data be replaced when the CEMS is not providing valid data, even if that source is not subject to the Acid Rain requirements. In those cases, the Division would develop a data replacement procedure.

The kiln is not subject to the Acid Rain requirements, nor was the kiln permitted as a synthetic minor source, therefore, the Division considers that data replacement is not required for the kiln CEMS. Note that while data replacement is not required, monitor downtime will be evaluated to ascertain whether the source is meeting the requirement to continuously operate their CEMS.

- The statement regarding the initial performance test for VOC emissions in Condition 10.7 has been removed from the permit, as it is no longer necessary.
- The language regarding performance test methods in Conditions 10.7 and 10.11 was revised to list the “appropriate EPA test method”, rather than a specific test method.

Various Sections - Miscellaneous Changes

Section II.5 – Raw Material Drying

- Removed the specific test methods in Condition 5.5 and noted the appropriate EPA test methods shall be used. This provides more flexibility in preparing and approving the test protocol.

Section II.11 – P009 thru P013 and P015

- The source asked for one grain loading limit for all of the baghouses but did not suggest a limit. As discussed later in this document, the Division lowered the grain-loading limits for many of the emission units within Section II.11 to levels that demonstrate compliance with the emission limitations. Therefore, most of the baghouses within each emission group have the same grain-loading limit.
- The source requested to revise the one-way haul distance to 0.1 mile. Since the underlying construction permit included a one-way haul distance of 0.3 mile, the suggested change will not be made.
- Replaced “possible” with “practicable” in Condition 11.5.6.

Section II.12 – P014 Railcar Unloading

- The source requested to use a different emission factor for materials other than coal or clinker. This change was made as requested.

Section II.13 – P007A – Handling and Processing of CKD and Raw Material Waste Dust

- The source requested that the daily throughput limit be revised to 640 tons/day for dry material. The request seems to be based on the daily throughput limit for wet material. The daily throughput and PM₁₀ limits were included in the underlying construction permit based on the results of a modeling analysis and thus should not

be increased unless additional modeling is conducted. Therefore, the requested change was not made.

- The hours of operation for S066 and S067 were revised to 8064 hours per year as requested. According to the April 7, 2004 Construction Permit (98BO0315), it appears that hours were limited to 8064 per year for the silos and loading spout. The technical review document for the first renewal (issued March 1, 2008) indicated that hours of operation for the waste dust silo were limited to 8064 hours per year and to 3000 hours per year for silo A5 and the loading spout (see page 5) but it appears that this statement was made in error.
- The source requested that the pressure drop monitoring for cement silo A5 be removed. According to the technical review document for the first renewal (issued March , 2008), this requirement was included because of concerns regarding the operation of the silo (pages 6 & 7). Note that pressure drop monitoring for cement silo A5 is also included in Section II, Condition 21. The frequency for monitoring the pressure drop for cement silo A5 (S066) was revised to weekly. Note that the requirement to monitor pressure drop for S066 was removed from Condition 21.
- The source requested that Condition 13.6.5 be revised to reflect that 3 acres may be used for disposal rather than 7, to be consistent with the COC requirement (Condition 13.6.13). Since Condition 13.6 5 addresses the same concerns as Condition 13.6.13 (which included the COC requirement), Condition 13.6.5 has been removed.

Section II.14 – P018 – General Fugitive Emissions Requirements

- The source requested that the following sentence be removed from Condition 14.4.1.2.c (truck wash requirements) “The permittee shall keep records of the time periods when the truck wash is not used and the reason it is not being used and shall make such records available to the Division upon request.” The sentence was not removed from the permit but was revised to only require that records be kept of periods when the truck wash is not used because it is non-operational.

August 4, 2014 Modification Application

Consent Decree (09-cv-0019-MSK-MEH) was filed April 19, 2013 and requires CEMEX to install SNCR on the kiln. The purpose of the August 4, 2014 modification application is for the installation of the SNCR system. The SNCR is intended to decrease NO_x emission, but it can also potentially result in an increase in condensable PM emissions.

In May 2013, CEMEX requested that approval to operate a portable SNCR system be provided to conduct internal testing as well as to conduct emissions testing to determine if installation of SNCR would result in an increase in PM emissions. The Division approved CEMEX’s request and requested that total (filterable plus condensable) PM, PM₁₀ and PM_{2.5} testing be conducted with and without the SNCR operating to determine if there is an increase in PM, PM₁₀ and PM_{2.5} emissions to see if an increase in

emissions are expected and whether the increase, if any, would exceed the significant level in Colorado Regulation No. 3, Part D, Section II.A.44.

Performance testing was conducted on the portable SNCR system in July 2013 and based on the results of the performance tests, the installation of SNCR would result in an increase in filterable PM (only filterable is considered for applicability) and total PM₁₀ emissions but the increases were below the significance level. Therefore, the Division instructed CEMEX to submit a minor modification to their Title V permit to address the installation of the SNCR.

An application was submitted to revise the Title V permit to address the installation of SNCR on August 4, 2014, which was re-submitted on August 7, 2014 to include the draft permit. After the submittal of additional information on September 5, 2014 and approval of the revised draft permit language on October 6, 2014, the application was determined to be complete.

Modification Type

The Division had indicated that the modification to install the SNCR would qualify as a minor modification and CEMEX submitted the application as a minor modification to the Title V permit. Colorado Regulation No. 3, Part C, Section X.A identifies those modifications that can be processed under the minor permit modification procedures. Specifically, minor permit modifications “are not otherwise required by the Division to be processed as a significant modification” (Colorado Regulation No. 3, Part C, Section X.A.6). The Division requires that “any change that causes a significant increase in emissions” be processed as a significant modification (Colorado Regulation No. 3, Part C, Section I.A.7.a). According to Part G of Regulation No. 3 (Section I.L, revisions adopted July 15, 1993, Subsection I.G for modifications) the Division considers that a significant increase in emissions is the potential to emit above the major stationary source significant level in Colorado Regulation No. 3, Part D, Section II.A.44.

The testing conducted on the portable SNCR system in July 2013 indicated an increase in filterable PM emissions and a decrease in condensable PM emissions. Note that this is based on the average of two runs with the SNCR on (ammonia on), since there was a sample bias detected in Run 1 with SNCR on (ammonia on) the condensable results from that run were excluded from the analysis. Based on the Division’s analysis of the July 2013 performance test there was no increase in total (filterable and condensable) PM_{2.5} emissions, an increase in total (filterable and condensable) PM₁₀ emissions and an increase in PM (filterable only). The increases and decreases are dependent on the basis of the results (i.e., lb/hr, lb/ton feed or lb/ton clinker). Based on the Division’s analysis, the increases in PM emissions using the most conservative values are shown in the table below. The ton/yr values were determined based on either the feed or production limits included in the permit (967,800 tons/yr of feed and 600,000 tons/yr of clinker).

	Emissions (tons/yr) ¹		
	Filterable PM ²	Total PM ₁₀	Total PM _{2.5}
SNCR On ³	96.0	140.31	53.22
SNCR Off	79.80	135.48	59.03
Change in Emissions	16.20	4.83	-5.81
PSD/NANSR Significance Level (T5 Minor Mod Level)	25	15	10

¹Emissions for filterable PM are based on the lb/ton clinker emissions rates and emissions for PM₁₀ and PM_{2.5} are based on the lb/ton feed emissions rates determined from the performance test.

²According to the definition of regulated NSR pollutant in Reg 3, Part D, Section II.A.40, condensable emissions are only included in PM₁₀ and PM_{2.5} for assessing applicability.

³There was a sample bias in the results of run 1 with SNCR on, which affected the condensable results. The results of run 1 (both filterable and condensable) were excluded in assessing emissions with the SNCR on.

Note that the July 2013 test report also indicated a potential increase in CO emissions. CEMEX submitted information on September 5, 2014 that indicates that the increase in emissions would be below the significance level, as indicated in the table below:

	CO Emissions (tons/yr)
Current Permitted (Potential) Emissions	396
Baseline Emissions (2007 – 2008) ¹	328
Change in Emissions	68
PSD/NANSR Significance Level (T5 Minor Mod Level)	100

¹Based on calendar year emissions for 2007 and 2008 reported on APENs (2007 CO = 311 tons/yr and 2008 CO = 345 tons/yr).

Modeling Requirements

As indicated previously, this project is anticipated to result in an increase in PM, PM₁₀ and possibly CO emissions (the analysis indicates that PM_{2.5} emissions will decrease). However, those emissions increases are below the modeling thresholds specified in the Division's Colorado Modeling Guideline's May 20, 2011 Updated Tables as indicated in the table below. Therefore, modeling is not warranted for this modification.

Pollutant	Modeling Threshold ¹		Project Emissions	
	Annual	Short-Term	Annual	Short-Term
PM ₁₀	15 tons/yr	82 lbs/day	4.83 tons/yr	28.8 lbs/day
CO	100 tons/yr	23 lbs/hr	68 tons/yr	16.9 lbs/hr

¹ The thresholds in the Modeling Guideline are not "bright-lines", i.e., modeling may be warranted for projects with emissions below the thresholds. However, for this project given the level of emissions this situation does not warrant modeling.

Discussion

Since the testing was conducted on a trial SNCR unit, the Division intended to require that performance testing be conducted after installation of the permanent SNCR to demonstrate that the installation of the unit will not result in an emission increase above the significance level. The required testing was expected to be similar to that performed for the trial SNCR, in that testing will be conducted with the SNCR off and the SNCR on to assess the change in emissions due to the SNCR.

The draft permit associated with the minor modification application (which was considered complete on October 6, 2014) included a requirement to conduct a performance test on the SNCR after installation. The performance test was tied to a report required by the Consent Decree (the SNCR optimization report) which according to CEMEX was not required to be submitted since CEMEX opted to take the default NO_x limit of 1.85 lb/ton clinker. EPA later confirmed to the Division that the NO_x optimization report would not be required.

When the Division determined that the SNCR performance test would not be triggered by the draft permit language, it was discovered that the baghouse bags had been replaced with membrane bags in order to meet the lower NESHAP PM limit that took effect in September 2015. Since the membrane bags would likely result in lower PM emissions from the kiln, the test would not be conducted under conditions similar to the trial SNCR test. In addition, an SO₂ scrubber will also be installed to meet the NESHAP HCl requirement, which would likely result in a decrease in condensable PM emissions.

Although the membrane bags and SO₂ scrubber may not have been intended to reduce or off-set any potential increases in PM due to the installation of SNCR, they are likely to mitigate the impacts of any potential increase in PM emissions due to the SNCR. Therefore the Division considers that a performance test on the permanent SNCR is no longer required and is not included in the renewal permit.

The source also indicated that the SNCR will install an aqueous ammonia tank to support the SNCR. In their September 5, 2014 information submittal, the source indicated that the aqueous ammonia would have an ammonia concentration no more than 19%, so the facility would not be subject to the provisions in 40 CFR Part 68, Chemical Accident Prevention Provisions.

Revisions to Permit

The following changes were made to the permit to address this modification:

Section I – General Activities and Summary

- The SNCR will be included in the Table in Condition 4.1, under “pollution control device” for the kiln.

Appendices

- The aqueous ammonia storage tank was included in the insignificant activity list in Appendix A.
- “New” appendix I was added to include the PSD/NANSR applicability test.

March 9, 2015 Modification Application

The purpose of the March 9, 2015 modification is to allow for the construction and operation of new equipment necessary to comply with the requirements in the Portland Cement MACT. CEMEX will be installing activated carbon and lime injection systems on the kiln to comply with the mercury (Hg) and hydrochloric acid (HCl) requirements in the Portland Cement MACT. The application indicates that the installation of the actual control devices on the kiln will not result in an increase in emissions from kiln and the Division agrees with this assessment. However, support equipment is required for the control devices and the purpose of this modification is to permit the proposed new support equipment.

The Division noted deficiencies in the March 9, 2015 application and a revised application was submitted on April 16, 2015.

The proposed new equipment for the kiln control devices is as follows:

- An activated carbon storage silo
- A lime storage silo
- A lime weigh hopper

All the proposed support equipment will be equipped with baghouses to control PM emissions.

Modification Type

The source indicated that this modification would qualify as a minor modification. Colorado Regulation No. 3, Part C, Section X.A identifies those modifications that can be processed under the minor permit modification procedures. Specifically, minor permit modifications “are not otherwise required by the Division to be processed as a significant modification” (Colorado Regulation No. 3, Part C, Section X.A.6). The Division requires that “any change that causes a significant increase in emissions” be processed as a significant modification (Colorado Regulation No. 3, Part C, Section I.A.7.a). According to Part G of Regulation No. 3 (Section I.L, revisions adopted July 15, 1993, Subsection I.G for modifications) the Division considers that a significant increase in emissions is the potential to emit above the major stationary source significant level in Colorado Regulation No. 3, Part D, Section II.A.44. Potential to emit is based on the maximum design rate, 8760 hours per year of operation and does not rely on controls, unless the source is subject to federally enforceable requirements that restrict operation or require controls. Potential to emit for these units is based on no controls, even though each unit will be equipped with a control device, maximum

estimated throughput rates, 8,064 hours per year of operation and emission factors from AP-42, Section 11.17. The kiln is subject to an operating limit of 8,064 hours per year of operation and since the equipment supports control equipment for the kiln, taking consideration for the operating limits on the kiln is appropriate. Estimated uncontrolled emissions from the project are as follows:

Emission Unit	Emissions (tons/yr)		
	PM	PM ₁₀	PM _{2.5}
Carbon Silo	0.06	0.06	0.06
Lime Silo	3.69	3.69	3.69
Lime Weigh Hopper	3.69	3.69	3.69
Total	7.44	7.44	7.44
PSD/NANSR Significance Level (T5 Minor Mod Level)	25	15	10

Since the potential to emit for this project is below the significance level, the Division agrees that this modification can be processed as a minor modification.

Discussion

Emission Factors

The emissions factor used to estimate uncontrolled emissions from the support equipment is 0.61 lb/ton and is from AP-42, Section 11.17, Table 11.17-4 (dated February 1998), for Product Unloading, enclosed truck. The uncontrolled emission calculations assume that PM = PM₁₀ = PM_{2.5} emissions.

Controlled emissions were estimated by applying a control efficiency of 99.9% for the baghouses.

Emission and Throughput Limits

Emissions (controlled and uncontrolled) from this project are shown in the table below.

Emission Unit ¹	Emissions (tons/yr)					
	Uncontrolled			Controlled		
	PM	PM ₁₀	PM _{2.5}	PM	PM ₁₀	PM _{2.5}
Lime Silo	3.69	3.69	3.69	3.69×10^{-3}	3.69×10^{-3}	3.69×10^{-3}
Lime Weigh Hopper	3.69	3.69	3.69	3.69×10^{-3}	3.69×10^{-3}	3.69×10^{-3}
Total	7.38	7.38	7.38	7.38×10^{-3}	7.38×10^{-3}	7.38×10^{-3}

¹Since uncontrolled emissions from the carbon silo are below the APEN de minimis level, the silo will be included in the insignificant activity list and is not shown here.

For the first year of operation, the support equipment shall be subject to monthly emission limits and those will be set at the annual limits divided by 12 (0.67 lbs/mo, each, for the lime silo and lime hopper based on an annual limit of 0.004 tpy).

Since the actual uncontrolled emissions for the activated carbon silo are below the APEN de minimis level a permit is not necessary for that silo.

The annual throughput is based on maximum hourly throughput rates of 3,000 lbs of lime per hour (12,096 tons/yr based on 8,064 hours per year of operation). For the first year of operation, the support equipment shall be subject to monthly throughput limits and those will be set at the annual throughput divided by 12 (1,008 tons/mo, each, for the lime silo and lime hopper).

Applicable Requirements

The applicable requirements for the lime silo and lime weigh feeder are as follows:

- Construction of these sources must commence within 18 months of initial approval permit issuance date or within 18 months of date on which such construction or activity was scheduled to commence as stated in the application (Reg 3, Part B, Section III.F.4.a.(i) thru (ii)).
- Within 180 days after commencement of operation, compliance with the conditions contained on this permit shall be demonstrated to the Division (Reg 3, Part B, Section III.G.2)

Note that the Division considers that the first semi-annual monitoring and permit deviation report submitted after the units commence operation will serve as the self-certification.

- The permittee shall notify the Division, in writing, fifteen (15) days after startup (Reg 3, Part B, Section III.G.1).
- 20% opacity (Regulation No. 1, Section II.A.1)

Based on engineering judgment, the Division has not included the 30% opacity requirement for startup, process modification and adjustment of control equipment (Reg 1, Section II.A.4) for the following reasons: 1) startup is instantaneous (begin loading or unloading); 2) process modifications are unlikely since the process of loading and unloading is straightforward and if modifications were to occur, they could not occur while the units are in operation (i.e. loading or unloading) and 3) the control equipment cannot be adjusted while loading or unloading is occurring.

- APEN reporting (Reg 3, Part A, Section II)

The APEN reporting requirements will not be identified in the permit as a specific condition but are included in Section IV (General Conditions) of the permit, condition 22.e.

- Lime processing **through each unit** not to exceed 12,096 tons/yr (based on information provided in the APEN received on March 9, 2015)
- PM, PM₁₀ and PM_{2.5} emissions **from each unit** not to exceed 0.004 tons/yr

(based on requested emissions provided in the APEN received on March 9, 2015)

The Division determined that neither the Regulation No. 1 (Section III.C.1) nor the Regulation No. 6 (Part B, Section III.C, including opacity) particulate matter standards were applicable to the lime silo or lime weigh hopper. The Division does not consider these to be manufacturing processes since the lime is used to control HCl emissions from the kiln.

Modeling Requirements

Emissions from this project are below the modeling thresholds specified in the Division's Colorado Modeling Guideline's May 20, 2011 Updated Tables as indicated in the table below. Therefore, modeling is not warranted for this modification.

Pollutant	Modeling Threshold ¹		Project Emissions	
	Annual	Short-Term	Annual	Short-Term
PM ₁₀	15 tons/yr	82 lbs/day	0.156 tons/yr	0.93 lbs/day
PM _{2.5}	5 tons/yr	11 lbs/day	0.156 tons/yr	0.93 lbs/day

¹ The thresholds in the Modeling Guideline are not "bright-lines", i.e., modeling may be warranted for projects with emissions below the thresholds. However, for this project given the level of emissions this situation does not warrant modeling.

Revisions to Permit

The following changes were made to the permit to address this modification:

Section I – General Activities and Summary

- The lime silo and lime weigh hopper was included in the Table in Condition 4.1. In addition, activated carbon and lime injection systems have been included in the Table in Condition 4.1 under "pollution control device" for the kiln.

"New" Section II.25 – Kiln Control Device Support Equipment

Note that in the current permit (last revised April 1, 2013) Section II.25 includes the requirements for the cement rail car unloading system.

- Provisions for the lime silo and lime weigh hopper have been included in this section.

Appendix A – Insignificant Activity list

- The activated carbon storage silo was included in the insignificant activity list.

Appendices B and C

- Included the lime silo and lime weigh hopper in the tables.

June 19, 2015 Modification Application

The purpose of the June 19, 2015 application is to increase the throughput limit for the Dowe Flats Quarry (“the quarry”). Following the Division’s review of the application, the source submitted additional information on July 24, 2015 and a revised application on August 4, 2015. The August 4, 2015 revised application requested that the throughput limit be increased at the quarry from 3,200,000 tons/yr to 3,500,000 tons/yr (the June 19, 2015 requested a throughput increase to 3,600,000 tons/yr). The source is not requesting an increase to any of the emission or throughput limits for raw materials used at the cement plant. The throughput increase is necessary because additional overburden needs to be removed to get to the raw materials necessary for the cement plant. The source has not requested an increase in the emission limitations for the quarry.

The June 19, 2015 application is essentially a re-submittal of an application submitted on February 13, 2014 and that the source requested to be cancelled on March 17, 2015.

Modification Type

The Division had indicated that the modification to increase throughput at the quarry would qualify as a minor modification and CEMEX submitted the application as a minor modification to the Title V permit. Colorado Regulation No. 3, Part C, Section X.A identifies those modifications that can be processed under the minor permit modification procedures. Specifically, minor permit modifications “are not otherwise required by the Division to be processed as a significant modification” (Colorado Regulation No. 3, Part C, Section X.A.6). The Division requires that “any change that causes a significant increase in emissions” be processed as a significant modification (Colorado Regulation No. 3, Part C, Section I.A.7.a). According to Part G of Regulation No. 3 (Section I.L, revisions adopted July 15, 1993, Subsection I.G for modifications) the Division considers that a significant increase in emissions is the potential to emit above the major stationary source significant level in Colorado Regulation No. 3, Part D, Section II.A.44.

The source is not requesting an increase in permitted emissions for the quarry with this modification. Since the facility is a major stationary source and permitted emissions from the quarry are above the significance levels, the source conducted an applicability analysis to demonstrate that there would not be a significant increase in emissions. This analysis is discussed in more detail later under “Discussion”. Therefore, since this modification does not result in a significant increase in emissions, the Division agrees that this modification qualifies for a minor modification.

In addition, Colorado Regulation No. 3, part C, Section X.A.4 specifies that those changes that “do not seek to establish or change a permit term or condition for which there is no corresponding applicable requirement and that the source has assumed to

avoid an applicable requirement to which the source would otherwise be subject” can be processed as a minor modification. This provision essentially prohibits sources from obtaining synthetic minor status with a Title V minor modification. The Dowe Flats Quarry was permitted as a “synthetic minor” source in the underlying construction permit that was incorporated into the Title V permit (93BO1414-4, mod 2, issued September 11, 1998) thus credit had already been taken to keep emissions below the significance level. Therefore the Division would consider that this modification would qualify as a minor modification as long as additional controls and/or a higher efficiency on existing controls were not utilized.

The current Title V permit does not specify emission factors in the permit for the quarry (the permit specified that compliance with the emission limits are presumed provided the throughput limits and control measures are met). Since the origin and underlying assumptions of the emission factors used in the construction permit and by the source to estimate emissions (for APEN purposes) was not entirely clear, for this modification the appropriate emission factors from AP-42 were used to estimate both baseline and projected actual emissions. Since the Title V permit did not include emission factors in the permit, it is more difficult to determine whether controls were anticipated when the quarry was permitted as a “synthetic minor”.

The June 19, 2015 application included control efficiencies for the following activities that were not previously used in emission calculations that have been submitted to the Division as part of the annual inspection and thus could be interpreted as “new” controls: drilling (90% control for baghouses), topsoil removal and unloading (50% for water), wind erosion (50% for and grading of haul roads (80% for water/chemical stabilizers). Although it appears that these activities were previously “uncontrolled” and thus would prevent the modification from qualifying as a minor modification, this is not the case.

For drilling, the construction permit listed the emission factors in the “notes to the permit holder” and specified that bag collectors are used but do not list a control efficiency for the drilling emission factors. The Title V permit specifies that drills shall be equipped with bag collectors to control emissions. Since the permit currently requires the bag collectors and 90% control is an acceptable control efficiency for a bag collector, the Division considers that taking 90% control does not represent a “new” control and does not prevent this modification from qualifying for a minor modification.

For grading of haul roads, the construction permit lists limited speed/watering in the emission factor table but does not list a control efficiency for grading of haul roads. The construction permit lists 80% control for hauling on the haul roads and while not specifically noted in the quarry section of the Title V permit, an 80% control for haul roads is listed in the permit and used by the source to calculate emissions from the quarry haul roads. Since an 80% control efficiency can be used for hauling on the haul roads (watering and chemical stabilizers are control measures specified in the Title V permit for the haul roads), it makes sense that an 80% control can be used for grading of those same haul roads. Thus the Division considers that this does not represent a

“new” control and does not prevent this modification from qualifying as a minor modification.

For topsoil loading, the source used 50% control efficiency for loading and unloading. The emission factors listed in the construction permit did not provide a control efficiency for loading and unloading of topsoil. Both the construction permit and the Title V permit include as control measures requirements to maintain adequate moisture in topsoil and overburden to minimize emissions and to control emissions from material handling (i.e. removal, loading, and handling) by watering at all times, unless natural moisture is sufficient to control emissions. Since the permit requires that watering of materials handled to control emissions, use of 50% control does not represent a “new” control and does not prevent this modification from qualifying as a minor modification.

For wind erosion, the source relied on 50% control for wind breaks. The quarry and exposed area are situated in a natural bowl depression that provides a wind break. Since the wind break is a natural barrier based on the location, the Division does not consider this to be a “new” control and does not prevent this modification from qualifying as a minor modification.

It should be noted that for drilling, grading of haul roads and wind erosion, the control efficiencies were used for both baseline and projected actual emissions, thus there was no advantage provided for the control efficiencies utilized. The activity of topsoil loading/removal was not included in the baseline actual emission calculations.

Modeling Requirements

As previously indicated, the source is not requesting an increase in permitted emissions from the quarry operations with this modification. Therefore, modeling is not required for this modification.

Discussion

The current Title V permit stipulates that compliance with the emission limits are met as long as the throughput limits and fugitive dust control measures are met. While this may have at one time seemed to be an appropriate method to assure compliance with the annual emission limitations, emission estimates are not strictly based on an emission factor times a throughput rate. For example, emissions from hauling materials are dependent not just on the throughput but the number of haul trips and length of the haul trips. The number of haul trips are dependent on the size of the trucks are how full the trucks are. Therefore, the Division considers that emission calculations should be conducted. In addition, a limit on the number of daily haul trips was included in the permit. the Division considers that a limit on the number of daily haul trips is appropriate as well as requirements to gather information to calculate emissions (such as vehicle miles traveled).

During the processing of the renewal permit, the Division realized that the projected emissions increases from the Dowe Flats Modification only addressed the increases in PM emissions from blasting but did not address the potential increase in combustion

emissions associated with blasting. Therefore in February 2016, the Division asked CEMEX for information related to combustion emissions from blasting. Since there would be an increase in emissions (projected increases indicated an increase in the number of blasts), emission limits for combustion emissions from blasting would be included in the permit. CEMEX submitted emission estimates from blasting on April 11, 2016. When the Division noted that the NO_x and CO emissions from blasting would be permit limits, the source submitted a revised APEN on August 25, 2016 requesting slightly higher emissions and throughput. Requested (permitted) emissions were below the significance level, therefore, major stationary source permitting requirements were not triggered and the modification still qualifies as a Title V minor modification. Requested (permitted) combustion emissions from blasting are as follows:

	Emissions (tons/yr)		
	NO _x	CO	SO ₂
Requested Emissions	10.0	39.6	1.2
Significance Level	40	100	40

Revisions to Permit

The following revisions were made to the permit to address this modification:

Section II.1 – Dowe Flats and Lyons Quarry – Fugitive Dust Sources

- Condition 1.2 was revised to include the emission factors for the Dowe Flats quarry operations and to require that emissions from quarry operations be calculated monthly and that 12 month rolling totals of emissions be maintained.
- “New” conditions were added to include the following:
 - NO_x and CO emission limitations from blasting
 - Requirements to maintain monthly records of days of operation, number of blasts, number of hours bulldozing occurred and number of vehicle miles traveled (VMT) and
 - A limitation on the number of daily haul trips.

Appendices

- The emission factors for the Dowe Flats quarry were included in “new” appendix “H”.
- “New” appendix I was added to include the PSD/NANSR applicability test.

August 21, 2015 Modification Application

The purpose of the August 21, 2015 application is to include the extension to the compliance date for the requirements in 40 CFR Part 63 Subpart LLL (PC MACT). The source received an extension to the PC MACT compliance date in a June 11, 2015 letter from the Division. The September 9, 2015 compliance date was extended by six

months thus making the compliance date March 9, 2016. The extension to the compliance date was granted for the mercury (Hg) and hydrochloric acid (HCl) requirements in order to install control equipment.

Modification Type

The August 21, 2015 application did not identify any specific modification procedures (e.g. minor modification), therefore the modification will be treated as a significant modification. A significant modification goes through the same process as a new or renewed Title V permit, which includes 30 day public comment and 45-day EPA review periods. Normally sources cannot institute the proposed modifications addressed in a significant modification application until the revised operating permit has been issued. However, the extension to the compliance date for the requirements in 40 CFR Part 63 Subpart LLL was granted in the Division's June 11, 2015 letter thus the extension to the compliance date is in place, regardless of whether a revised permit containing the conditions of the June 11, 2015 letter has been issued.

Modeling Requirements

This modification does not result in an increase in emissions, therefore modeling is not required.

Revisions to Permit

This document and the renewal permit are still in process after the extended compliance date for the HCl and Hg PC MACT limits (March 9, 2016). The HCl and Hg controls have been installed on the kiln and the source is in the process of conducting the initial compliance tests (initially scheduled to occur in late April or early May of 2016). Therefore, the conditions upon which the PC MACT compliance date was extended have passed and thus do not need to be included in the permit. However, a note will be included in the PC MACT section of the permit (Section II.23) that the compliance date was extended for the Hg and HCl limits.

October 30, 2015 Modification Application

The purpose of the October 30, 2015 application is to include the relevant requirements from the CD (09-cv-0019-MSK-MEH) into the Title V permit.

On October 30, 2015 CEMEX submitted information identifying "SNCR requirements for Operating Permit". The submittal identified the final NO_x limit, as well as monitoring requirements that were to be included in a permit as required by the CD but did not specifically state that they were applying for a permit revision. When asked, CEMEX indicated that the October 30, 2015 submittal was intended to fulfill the requirement in the CD to incorporate certain CD requirements into the Title V permit.

Paragraph 30 of the CD specifically states that CEMEX shall apply to the Division to include certain CD requirements into "a federally enforceable permit (other than a Title V permit)" and then within 60 days of issuance of the federally enforceable permit,

CEMEX would be required to submit an application to incorporate the requirements of that permit into the Title V permit (per paragraph 31 of the CD). In an email dated April 1, 2016, the Division obtained approval from EPA to incorporate the CD requirements directly into the Title V permit.

Modification Type

The October 30, 2015 application did not identify any specific modification procedures (e.g. minor modification), therefore the modification will be treated as a significant modification. A significant modification goes through the same process as a new or renewed Title V permit, which includes 30 day public comment and 45-day EPA review periods. Normally sources cannot institute the proposed modifications addressed in a significant modification application until the revised operating permit has been issued. However, CEMEX is required to comply with the CD NO_x limits and monitoring and SNCR operating requirements, regardless of whether a revised permit containing the CD requirements has been issued

Modeling Requirements

This modification does not result in an increase in emissions, therefore modeling is not required.

Discussion

Paragraph 30 of the CD specifies that the following requirements from the CD are to be included in a federally enforceable permit: the final NO_x limit, the operational requirements for the SNCR system, the monitoring requirements in the CD and the requirements in Section VII of the CD (prohibition on netting credits or offsets). The application did not address the requirements in Section VII of the CD nevertheless these requirements will be included in the permit.

Revisions to Permit

The following revisions were made to the permit to address the October 30, 2015 modification:

Section II.10 – Kiln

- The Consent Decree NO_x limit was included in “new” Condition 10.10.
- The SNCR operating requirements were included in “new” Condition 10.20.
- The prohibitions on netting credits or offsets from required controls were included in “new” Condition 10.21.

September 26, 2016 Comments on the Draft Permit and Technical Review Document

Based on the source's September 26, 2016 comments on the draft permit and technical review document, the following changes were made to the permit:

Note: The references to revised permit conditions in this particular section is with respect to the draft renewal permit, not the current permit (last revised April 1, 2013) as it is in the other sections of the document.

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- The addresses under "Issued to" and "plant site location" were revised.

Section I – General Activities and Summary

- Revisions to equipment description or control method were made to the haul roads (P018, AIRS pt 019), raw material separator (S011, AIRS pt 005), outdoor clinker storage (P015, AIRS pt 015) and the masonry storage silos (S043, AIRS pt 013) in the summary table in Condition 4.1. The description revisions were also made in Condition 6.1 (CAM) for S011 and S043.
- The north and south packer (S049 and S050, AIRS pt 013) were removed from the summary table in Condition 4.1 since the equipment is inoperable (the power system from each packer has been removed). This equipment was also removed from Condition 6.1 (CAM).

Section II.1 – P017 – Dowe Flats and Lyons Quarry – Fugitive Dust Sources

- Condition 1.6.1.3 was revised to allow for the use of a water truck or sprinkler system (the current language specifies that a sprinkler system be used).
- Condition 1.6.1.12 (monitor and record local weather forecast information) was removed since this requirement from the 2002 COC was not required to be included in the Title V permit.

Regulation No. 1 PM Limitation (various sections)

- The language for the Reg 1 PM limit (manufacturing processes) was revised to be consistent with the language in the regulation. This affected Conditions 4.3, 6.3, 7.3, 8.3, 9.3 and 12.3

Section II.3 – Raw Material Storage and Handling

- The "iron slag" in listed in the summary table in Condition 3.1 and as a footnote was revised to "iron containing material."

Section II.5 – Raw Materials Drying

- Condition 5.7.1 was revised to indicate that the last performance test was conducted in June 2016 and Condition 5,7,2 was revised to include the emission factors from the June 2016 performance test.

Section II.11 – P009, P010, P015, P011 and P012 and P013 – Various Material Handling Sources

- The north and south packer (S049 and S050, AIRS pt 013) were removed from the summary table in Section II.11 since the equipment is inoperable (the power system from each packer has been removed).
- The north and south packer (A049 and S050, AIRS pt 013) baghouses (925-1 and 925-2) were removed from the table in Condition 11.4.2 and the grain-loading specifications were revised to reflect the removal of these baghouses.
- The north and south packer (A049 and S050, AIRS pt 013) were removed from Condition 11.8 (CAM requirements).
- Condition 11.6.1.2 (water truck requirement) was revised to specify a water application system (such as a water truck or sprinkler system). Specific requirements were included for the sprinkler system similar to the requirements for the CKD disposal area (Condition 13.7.1.8 in the draft renewal permit). In addition, the language for the water truck requirements was revised to be consistent with the language in Condition 14.4.1.2 in the draft renewal permit.

Section II.14 – General Fugitive Emission Requirements

- Condition 14.4.1.2 was revised to allow for a water truck or a sprinkler system. Specific requirements were included for the sprinkler system similar to the requirements for the CKD disposal area (Condition 13.7.1.8 in the draft renewal permit). In addition, the Condition 14.4.1.2.a was revised to remove the requirement that operation of the water truck shall be the sole assignment of the individual and to remove the specific measure to prevent freezing conditions.

Section II.23 – CAM Requirements

- The following changes were made to the summary table in Condition 23:
 - The equipment descriptions for the raw material separator (S011, AIRS pt 005) and the masonry storage silos (S043, AIRS pt 013) were revised.
 - The north and south packer (S049 and S050, AIRS pt 013) were removed.

Appendices B and C

- The equipment description for outdoor clinker storage (P015) was revised.

Appendix G – CAM Plan

- The equipment descriptions for the raw material separator (S011, AIRS pt 005) and the masonry storage silos (S043, AIRS pt 013) were revised.
- The north and south packer (S049 and S050, AIRS pt 013) were removed.

November 21, 2016 Comments on the Draft Permit and Technical Review Document

Based on the source's November 21, 2016 comments on the draft permit and technical review document, the following changes were made to the permit:

Note: The references to revised permit conditions in this particular section is with respect to the draft renewal permit, not the current permit (last revised April 1, 2013) as it is in the other sections of the document.

Section I.5 – Alternative Operating Scenarios

- An alternative operating scenario was added to allow the source to use the MACT LLL THC compliance option for the dryer and the total organic HAP option for the kiln.

In their pre-public comment review comments (received September 25, 2016) the source asked the additional MACT LLL compliance options be added for the kiln and dryer. The Division's response noted that changes would necessary for other compliance options, such as installing a new monitoring system thus changes to compliance options could be appropriately addressed through a minor modification.

In their comments submitted during the public comment period, the source noted that for the dryer, no additional equipment needed to be installed in order to switch from the organic HAP option to the THC option. The Division's preliminary response to this comment was that the source has not been able to demonstrate compliance with the THC option, thus the option should not be provided. The source noted that future changes to the raw material source or quarry location may make the site able to comply with the THC limit for the dryer without controls. As a result the Division included the THC option as an alternative operating scenario. The source will be required to submit a notification indicating the intent to change compliance options and to describe changes to operations, such as installation of controls or changes to the raw material source or quarry location that that will ensure the dryer can comply with the THC limit.

Although the source's comments appeared to agree that since additional equipment would be required for other compliance options for the kiln, a permit modification would be appropriate for those options. Nevertheless, the Division included the total organic HAP option for the kiln as an alternative operating scenario, since this option

utilizes the THC monitor and the provisions for the total organic HAP option are included in the permit.

Section II.3 – Raw Material Storage and Handling

- Revised Condition 3.5.1.6 to specify “high efficiency industrial sweeper”.

Section II.11 – P009, P010, P015, P011 and P012 and P013 – Various Material Handling Sources

- Revised Condition 11.6.1.9 to specify “high efficiency industrial sweeper”.

Section II.14 – General Fugitive Emission Requirements

- Revised the language in Condition 14.4.1.1 to allow the use of any chemical dust stabilizer or dust suppressants to control fugitive particulate matter emissions.

Other Modifications

In addition to the source requested modifications, the Division has included changes to make the permit more consistent with recently issued permits, include comments made by EPA on other Operating Permits, as well as correct errors or omissions identified during inspections and/or discrepancies identified during review of this renewal.

The Division has made the following revisions, based on recent internal permit processing decisions and EPA comments to the Lyons Cement Plant Renewal Operating Permit. These changes are as follows:

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- Monitoring and compliance periods and report and certification due dates are shown as examples. The appropriate monitoring and compliance periods and report and certification due dates will be filled in after permit issuance and will be based on permit issuance date. Note that the source may request to keep the same monitoring and compliance periods and report and certification due dates as were provided in the original permit. However, it should be noted that with this option, depending on the permit issuance date, the first monitoring period and compliance period may be short (i.e. less than 6 months and less than 1 year).

Section I – General Activities and Summary

- The consent decree (09-cv-0019-MSK-ME) filed on April 19, 2013 was included in Condition 1.3 as a source of applicable requirements.
- Added a statement to Condition 1.5 indicating that either electronic or hard copy records are acceptable.

- Revised the reg citations for the definitions of “net emissions increase” and “significant” in Conditions 2.1 and 2.2. These citations were revised due to revisions to Reg 3.
- The diesel and natural gas-fired engines located at the facility were added to the table of emission units in Condition 4.1.

Section II - General

- Many of the permit conditions specifically state that records be kept of the daily, monthly and twelve month rolling totals and made available to the Division upon request. The recordkeeping requirements in the general permit conditions (Section IV.22) require that records of required monitoring be retained (see Section IV.22.b & c) thus is it not necessary to repeat the requirement to retain records in permit conditions within Section II.
- The provisions for bagfilter operation and maintenance requirements that are found in several sections in Section II have been included in “new” Condition 20. Note that Condition 20 in the current permit (last revised April 1, 2013) includes NSPS OOO requirements.
- The permit typically uses the terms “bagfilters” and “baghouses”. In order to use consistent terminology in the permit was revised to use the term “baghouse”.

Section II.1 – Dowe Flats and Lyons Quarry – Fugitive Dust Sources

- Allowable PM and PM₁₀ limits for disturbed areas at the Lyons quarry were included as a separate condition and language was added to require that emissions from the Lyons quarry be estimated.
- Condition 1.3.6 was split to clarify that there are requirements for watering and spill cleanup.
- The fugitive particulate matter requirements in Condition 1.4 were re-organized and revised to more appropriately reflect the requirements in Colorado Regulation No. 1 that apply to these sources.

Section II.2 – Dowe Flats Quarry – Point Source Emissions

- The NSPS Subpart OOO requirements (included in Condition 20 in the current permit (last revised April 1, 2013) have been moved to Condition 2.2, since only the Dowe Flats Quarry - Point Source Emissions (Section II.2) are subject to these requirements. The following changes to the NSPS OOO requirements were also made:
 - The requirement to submit copies of Method 9 that indicate exceedances of the opacity limit with the next semi-annual report have been removed.

- Added language to the opacity semi-annual method 9s and the 5 year performance test requirement indicating that a different conveyor baghouse has to be tested for each event, until all baghouses have been tested then to repeat the cycle.
- Added language to indicate that the opacity semi-annual method 9s shall be separated by at least four (4) months.
- The performance test requirement was included as “new” Condition 2.5 and requires testing for the annual (tons/yr) limit as well as the NSPS OOO requirement.

Since initial issuance, the Title V permit (issued February 1, 2000) has required performance tests to monitor compliance with the NSPS OOO limits. The Division considers that since PM testing is being conducted for the NSPS OOO limits, it should also be conducted for the annual PM limits.

The performance test language includes the following:

- The language specifies that a different conveyor baghouse has to be tested for each event, until all baghouses have been tested after which the cycle is repeated
- The language indicates how the test results are to be used to assess compliance with the annual (tons/yr) limits.
- The language requires that the test results be converted to lb/ton and compared the emission factors. If the test results indicate a higher emission factor than included in the permit, a permit application must be submitted to revise the emission factor.
- The NSPS General Provisions were included in “new” Condition 2.6. In the current permit some of the general provisions were included in Condition 20 and the good practices requirement was included in Condition 19. Since only the Dowe Flats Quarry - Point Source Emissions (Section II.2) refer to Condition 19, the NSPS general provisions were included in “new” Condition 2.6.

Section II.3 – Raw Material Storage and Handling

- Most of the language in the second paragraph of Condition 3.1 (throughput limit) was removed. The requirement to record days of operation will be included as a “new” separate condition.
- The daily throughput limit in Condition 3.1 was revised downward to reflect the emission factor and allowable daily PM₁₀ limit. The discussion of the January 9, 2002 modification of the Title V permit included in Appendix F of the current permit (last revised April 1, 2013), indicates that throughput and emission in Section II.3 were revised per construction permit modifications and that the throughput limit was corrected from the construction permit to reflect the actual requested throughput upon which the emissions were based. It appears that this change was correct for

the January 9, 2002 revised permit (based on the PM₁₀ emission factor and daily throughput limit, the daily PM₁₀ limit was met).

However, when the underlying construction permit (98BO0292) was revised in 2006, the emission factors were revised (increased) but the daily throughput and PM₁₀ limits were not changed. As a result, based on the daily processing rate in the current permit (last revised April 1, 2013) and the PM₁₀ emission factor, the PM₁₀ limit would not be met. Therefore, the daily throughput limit was reduced.

- Added a “new” separate condition to record days of operation. In the current permit (last revised April, 2013), the requirement to record days of operation is included in Condition 3.1 (throughput limit).
- Added language to Condition 3.2 (emission limitations) to indicate that daily emissions will be estimated by dividing monthly emissions by the number of days the equipment operated.
- Clarified the requirements in the summary table under Condition 3.3 (Reg 1 opacity requirements).
- The language in Condition 3.4.5 regarding ceasing operation of front-end loaders and hauling under certain wind conditions was removed since Condition 3.4.7 includes essentially the same language.
- The statement in the paragraph following Condition 3.4.7 indicating that the annual certification shall include a statement that the activities in Condition 3.4.7 were suspended when the average wind speeds were 30 mph or higher was removed. Typically the Division does not require specific language in the certification. The annual certification itself, signed by the Responsible Official is considered sufficient.
- The fugitive particulate matter requirements in Condition 3.4 were re-organized and revised to more appropriately reflect the requirements in Colorado Regulation No. 1 that apply to these sources.

Section II.4 – P001 – Primary Crusher (Plant Site)

- Revisions were made to the language in Conditions 4.1 and 4.2.
- Clarified the requirements in the summary table under Condition 4.4 (Reg 1 opacity requirements).

Section II.5 – Raw Materials Drying

- The hours of operation and dryer feed were included in separate conditions.
- A “new” condition was added to determine the Btu content of the fuel. This requirement was previously included in Condition 5.3 (dryer heat input limit).

- Include the RH requirements in the permit.

Note that the RH limits are the same as the limits included in the current Title V permit for NO_x, SO₂ and PM (which came from construction permit 12BO444-1). The construction permit PM limits were streamlined since they are the same as the RH limits. The RH opacity limit (10%) was streamlined in favor to the NSPS F opacity limits (less than 10%). Since the compliance date for the NO_x and SO₂ limits has not passed, both the construction permit and regional haze limits have been included in the permit. The RH NO_x and SO₂ limits are included in “new” Condition 5.13.

The RH monitoring requirements were included in the permit for NO_x, SO₂ and PM (performance tests, maintaining 12 month rolling totals and CAM (for PM)).

Note that since the requirements to maintain records for Subpart LLL (Section VII.D.3) and to submit semi-annual reports for CAM excursions and performance tests within 60 days (Section VII.E) are already addressed in the permit, the RH requirements will be streamlined (note that Reg 3, Part F, Section VII.E only required submittal of test reports for PM).

The requirements in Section VII.D.1 and 2 apply to CEMS and thus don’t apply to the dryer. In addition the requirement to submit EERs in Section VII.E also does not apply to the dryer since CEMS are not used.

- The frequency of performance testing for VOC (Condition 5.5) was revised to 30 months, rather than every 5 years, to tie the VOC testing with the testing for o-HAP required under NESHAP LLL.
- Language was added to the various performance test requirements to indicate how the results of the tests would be used to assess compliance with the emission limitations.
- Clarified the requirements in the summary table under Condition 5.6 (Reg 1 opacity requirements).
- The opacity requirement in Condition 5.8 (MACT opacity standard) was revised to include the NSPS Subpart F opacity requirement.
- A “new” condition was added for the NESHAP LLL requirements. Note that while the dryer is subject to the more stringent opacity requirement in NSPS F, it is still subject to the organic HAP and work practice requirements and the operation and maintenance plan requirements, as well as any related testing, monitoring, recordkeeping and reporting requirements in NESHAP LLL.

Section II.6 – P003 – Secondary Crushing

- Revisions were made to the language in Conditions 6.1 and 6.2.

- Clarified the requirements in the summary table under Condition 6.4 (Reg 1 opacity requirements).

Section II.7 – P004 – Raw Material Storage Silos

- Revisions were made to the language in Conditions 7.1 and 7.2.
- Clarified the requirements in the summary table under Condition 7.4 (Reg 1 opacity requirements).
- Clarified the NESHAP LLL requirements in Condition 7.5. The language in the table noting that a Method 9 test is done every five year has been removed. This requirement is not longer in h NESHAP LLL.

Section II.8 – P005 – Raw Material Grinding

- Revisions were made to the language in Conditions 8.1 and 8.2.
- Clarified the requirements in the summary table under Condition 8.4 (Reg 1 opacity requirements).
- Clarified the NESHAP LLL requirements in Condition 8.5. The language in the table noting that a Method 9 test is done every five year has been removed. This requirement is not longer in h NESHAP LLL.
- Added a “new” Condition 8.6 for CAM.

Section II.9 – P006 – Homogenizing and Blending

- Revisions were made to the language in Conditions 9.1 and 9.2.
- Clarified the requirements in the summary table under Condition 9.4 (Reg 1 opacity requirements).
- Clarified the NESHAP LLL requirements in Condition 9.5. The language in the table noting that a Method 9 test is done every five year has been removed. This requirement is not longer in h NESHAP LLL.

Section II.10 – P007 Kiln Burning

- In general, the statements specifying that records be kept (e.g. daily kiln feed rate) were removed, since General Condition 22 stipulates that records be kept and the duration for maintaining those records.
- The order was changed for some permit conditions and some permit conditions were separated and/or grouped together.

- Added language to Condition 10.2 (hours of operation) to clarify that annual hours of operation are on a calendar year basis.
- The kiln feed and fuel requirements were split into separate conditions.
- Added language to the SO₂, NO_x and CO emission limit conditions to indicate how lb/hr emission rates are calculated using the CEMS and to indicate that replaced data is used to assess compliance with the annual limitations.
- A separate condition was included in this section for the CEMS/COMS (“new” Condition 10.15)
- The PM emission limit (0.275 lb/ton dry feed) in Condition 10.4 is less stringent than the PC MACT PM emission limitation (0.07 lb/ton clinker), therefore it was streamlined from the permit (included in the permit shield for streamlined conditions, Section III.3).

Note that although the emission limitations are in different units, on a mass basis, the PC MACT limit is more stringent (21 tons/yr vs. 133 tons/yr). Assuming a conversion of feed to clinker of 0.64, based on the short term emission rate, the PC MACT limit is still more stringent (5.4 lb/hr vs 33 lb/hr).

- The Reg 1 process weight rate PM limits are included for S017 and S023 (sources under the P008 group). Note that while the Reg 1 process weight rate limits apply to the kiln (P007/S016) and clinker cooler (S018), the PC MACT PM requirements are more stringent thus the Reg 1 PM limits have been streamlined.
- A requirement (“new” Condition 10.8) was added to address specific requirements for burning tire-derived fuel. This includes the performance testing requirement in Condition 10.4, as well as the requirement to use tires that are managed under the oversight of a tire collection program.
- The requirement in Condition 10.5 (20% opacity requirement) to submit copies of the blasting event date and time information, the Method 9 observations and the COM data during the blasting event on a quarterly basis was removed. The Division considers that submitting this information is no longer necessary. The Division can review this information on site during an inspection.
- The requirement to determine HCl emissions annually for purposes of APEN reporting (Condition 10.10) has been removed. In practice, the Division has generally not included specific requirements in Title V permits to estimate emissions from HAPs for purposes of APEN reporting only. The requirement to file revised APENs is included in General Condition 22.e. Typically calculating HAP emissions has only been required if the source is subject to a ton per year HAP limit (via a construction or Title V permit requirement) or if the source close to a threshold and thus tracking HAP emissions from insignificant activities. Since the source is not subject to a ton per year HAP limit, the requirement to calculate HCl emissions has

been removed from the permit. Note that if HCl emissions are above the reportable level, they must be reported on any revised APENs that are submitted.

- The RH requirements were included in “new” Condition 10.22.

The kiln is subject to RH emission limitations for NO_x, SO₂ and PM. The RH PM emission limit is the same as the limit in Condition 10.4 and as discussed above, that limit is less stringent than the PC MACT limit. Therefore, the PM RH limit has been streamlined.

The kiln is also subject to a RH 20% opacity requirement and is discussed in further detail below.

The RH SO₂ and NO_x ton/yr limits are more stringent than the construction permit limits (12BP444-2). However, since the compliance date for the NO_x and SO₂ limits has not passed, both the construction permit and regional haze limits have been included in the permit. The RH NO_x and SO₂ limits are included in “new” Condition 10.22.

The RH PM monitoring requirements for the kiln intended to rely on a PM CEMS (the September 9, 2010 revisions to the PC MACT required that sources use a PM CEMS to monitor compliance with the PM limits) but noted that if the PC MACT was revised, stayed or vacated such that a PM CEMS was not required that the compliance with the PM limits would be monitored via performance tests. The frequency for conducting performance tests under the RH requirements is annual, although if the test results are less than 50% or 75% of the standard, the test frequency may be less frequent. Since the PC MACT relies on annual tests and monitoring using a PM CPMS, the monitoring for the RH PM limits is also less stringent and has been streamlined.

In addition to the performance tests, compliance with the PM limit would also rely on the CAM requirements. Since the CAM monitoring for the kiln, relies on the PC MACT monitoring, this is as stringent as the PC MACT monitoring, so this requirement will be streamlined.

Under the RH requirements, the RH 20% opacity requirement does not apply if a PM CEMS is used to monitor compliance with the PM limit (see Regulation No. 3, Part F, Section VII.C.2.a). The PC MACT was revised on February 12, 2013 to require performance testing in conjunction with a PM CPMS to monitor compliance with the PM limit. Under the February 12, 2013 revisions kilns were still not subject to an opacity limit.

Under the RH requirements, if a PM CEMS is not required, then the opacity limit applies and a continuous opacity monitoring system (COMS) shall be used. Since the PC MACT no longer requires a PM CEMS but relies on performance tests and a PM CPMS, the RH opacity requirement does apply and a COMS shall be used. The RH requirements do not specify whether the 20% opacity requirement applies at all

times. Absent any information in the RH requirements that provides exceptions from the opacity requirement, the Division assumes that this RH opacity limit applies at all times. Therefore, the RH opacity requirement has been included in the permit.

Note that since the record keeping requirements in Section VII.D and reporting requirements in Section VII.E have also been streamlined, since there are similar requirements already addressed in the permit.

Section II.11 – P009, P010, P015, P011 and P012 and P013 – Various Material Handling Sources

- Most of the language in the second paragraph of Condition 11.1 (throughput limit) was removed. The requirement to record days of operation will be included as a “new” separate condition.
- Added a “new” separate condition to record days of operation. In the current permit (last revised April 1, 2013) the requirement to record days of operating is included in Condition 11.1 (throughput limit).
- The grain loading limits/emission factors and the design rate of the baghouses specified in the Condition 11.3 of the permit do not ensure compliance with the annual PM and PM₁₀ emission limitations. Using the grain-loading specifications, the design rate of the baghouse (converted from acfm to dcfm) and allowable hours of operation, the emission limits would be exceeded for each emission group. Therefore, the grain-loading limits/emission factors were revised to ensure compliance with the annual PM and PM₁₀ emission limitations and the daily PM₁₀ emission limitations. Past performance test results indicate that the actual grain loading limits/emission factors are generally much less than those specified in the permit, therefore, this change is not expected to result in compliance issues.

The revised grain-loading limits/emission factors for P013 are based on the removed of the south (S049) and north (S050) packers. In their comments on the draft permit submitted on September 26, 2016, the source requested that this equipment be removed from the permit since it has been rendered inoperable.

- Condition 11.3 (emission limits) was reformatted and re-arranged to clarify the requirements.
- Language was added to the performance test requirement to indicate how test results would be used to assess compliance with the annual emission limitations.
- The permit (Condition 11.3) was revised to require that emissions be calculated monthly for P015. In addition, requirements were added to maintain records of vehicle miles travelled.
- The fugitive particulate matter requirements in Condition 11.5 were re-organized and revised to more appropriately reflect the requirements in Colorado Regulation No. 1

that apply to these sources. In addition, the following changes to existing control measures were made:

- The opacity requirement in Condition 11.5.1 was removed and replaced with visible emissions. The opacity requirements in 40 CFR Part 63 Subpart LLL do not apply to storage “piles”, although revisions to Subpart LLL include requirements for outdoor clinker storage piles.
- The control measure for treating haul roads with a chemical dust suppressant was grouped with a measure to operate a water truck in Condition 11.5.2. These conditions were separated to provide more clarity.
- Added language to Condition 11.5.2 (water truck requirement) to indicate areas that are subject to watering requirements (e.g. roads and other traffic areas, loading areas and other sources of fugitive particulate matter emissions).
- Clarified the requirements in the summary table under Condition 11.4 (Reg 1 opacity requirements).
- The opacity requirement in Condition 11.6 was revised to include the NSPS Subpart F opacity requirement.
- Added a “new” Condition 11.8 for the NESHAP LLL requirements. Note that while these sources are subject to the more stringent opacity requirement in NSPS F, they are still subject to the operation and maintenance plan and outdoor clinker storage requirements, as well as any related recordkeeping and reporting requirements in NESHAP LLL.

Section II.12 – Material Handling Systems – Load-In and Load-Out

- Revisions were made to the language in Conditions 12.1 and 12.2.
- Clarified the requirements in the summary table under Condition 12.4 (Reg 1 opacity requirements).
- Clarified the NESHAP LLL requirements in Condition 12.5. The language in the table noting that a Method 9 test is done every five year has been removed. This requirement is not longer in h NESHAP LLL.

Section II.13 – P007A – Handling and Processing of CKD and Raw Material Waste Dust

- Most of the language in the second paragraph of Condition 13.1 (throughput limit) was removed. The requirement to record days of operation will be included as a “new” separate condition.
- Added a “new” separate condition to record days of operation. In the current permit (last revised April 1, 2013) the requirement to record days of operation is included in Condition 13.1 (throughput limit).

- The permit (Condition 13.2) was revised to require that emissions be calculated monthly for hauling and disposal (042). In addition, requirements were added to maintain records of vehicle miles travelled.
- Condition 13.2 (emission limits) was reformatted and re-arranged to clarify the requirements.
- Language was added to the performance test requirement to indicate how test results would be used to assess compliance with the annual emission limitations.
- The fugitive particulate matter requirements in Condition 13.6 were re-organized and revised to more appropriately reflect the requirements in Colorado Regulation No. 1 that apply to these sources. In addition, the following changes to existing control measures were made:
 - The control measure for daily compliance inspections (Condition 13.6.9) was moved (no longer grouped as a “control measure”) and language was added to be more consistent with the weekly inspections required for other fugitive particulate matter sources.
 - The control measures for the water truck (Condition 13.6.3) and operating a sprinkler system (Condition 13.6.6) were removed, as there are more specific requirements on water truck and sprinkler system operation in Conditions 13.6.11 and 13.6.12.
 - The control measure for treating haul roads with a chemical dust suppressant was included as a separate control measure (it was grouped with a measure to operate a water truck in Condition 13.6.3).
- Clarified the requirements in the summary table under Condition 13.5 (Reg 1 opacity requirements).
- The opacity requirement in Condition 13.4 was revised to include the NSPS Subpart F opacity requirement.
- Added a “new” Condition 13.9 for the NESHAP LLL requirements. Note that while these sources are subject to the more stringent opacity requirement in NSPS F, they are still subject to the operation and maintenance plan, as well as any related recordkeeping and reporting requirements in NESHAP LLL.

Section II.14 – General Fugitive Emissions Requirements

- The requirements were reorganized and revised to more appropriately reflect the fugitive particulate matter requirements in Colorado Regulation No. 1.
- Condition 14.4.1.2.c was renumbered to 14.4.1.3 and the language in Condition 14.4.1.2 was revised to removed the phrase “and truck wash”. These changes were made as the requirements to use a truck wash was not a COC requirement but was a requirement resulting from local concerns.

Section II.16 – Opacity – Daily Plant Walk Through

- The title of this section was revised. In addition, the language in this condition was reformatted, revised and condition numbers were added to clarify the requirements.
- Added language indicating when daily visible emission observations are not required. These situations are when operations are not occurring or when operations are not operating for four or more consecutive daylight hours.

Section II.17 – Insignificant Activities

- This condition was removed, since the Division has generally not included such a requirement in Title V permits. It should be noted that insignificant activities are subject to opacity requirements and are included on the semi-annual monitoring reports and annual compliance certifications. In addition source's are required to maintain records verifying that certain activities qualify as insignificant.

“New” Section II.18 – Continuous Emission Monitoring and Continuous Opacity Monitoring Systems

- Requirements for the continuous emission and opacity monitoring systems were included in this condition. Note that in the current permit (last revised April 1, 2013) Section II.18 includes requirements for cold cleaner solvent vats.

“New” Section II.19 – Baghouse Operation and Maintenance

- Requirements for baghouse operation and maintenance were included in this condition. Note that in the current permit (last revised April 1, 2013) Section II.19 includes the good operating practices requirements. Baghouse operation and maintenance requirements are included throughout the permit but are being included in a centralized location in the permit to minimize repetition of the specific language. The equipment specific sections will refer to this Section II.20 for baghouse operation and maintenance requirements.
- Based on EPA's response to a petition on another Title V operating permit, language was added to require that records be maintained since this condition relies on “good engineering practices” (GEP).

Section II.21 – Opacity Limits

These requirements were moved to “new” Section II.20, the below discussion addresses the requirements as they appear in the current Title V permit (last revised April 1, 2013).

- The title of this section was revised to indicate that Condition 21 includes the Regulation No. 1 opacity requirements. Note that in the current permit (last revised April 1, 2013) Section II.20 includes the NSPS OOO requirements.

- The requirements in this condition were reformatted to clarify requirements.
- For the 30% opacity limit, the requirement to conduct additional observations if any observation indicates an exceedance of the limit was removed, as the language following that statement addresses that issue (violation persists until there is a opacity observation that indicates compliance).
- For the method 9 observations required for the 30% opacity limit, the requirement to submit copies of Method 9 observations that indicate exceedances of the opacity limit with the next semi-annual report have been removed.

Section II.22 – PM Stack Testing

These requirements were moved to “new” Section II.21, the below discussion addresses the requirements as they appear in the current Title V permit (last revised April 1, 2013).

- The performance test language was updated to the latest version.

Section II.23 – NESHAP LLL

These requirements were moved to “new” Section II.22, the below discussion addresses the requirements as they appear in the current Title V permit (last revised April 1, 2013).

- The summary table was removed from this condition.

Section II.25 – Cement Rail Car Unloading System

These requirements were moved to “new” Section II.24, the below discussion addresses the requirements as they appear in the current Title V permit (last revised April 1, 2013).

- Added language to Condition 25.1 to indicate that any information used to determine the monthly quantities of cement processed shall be maintained and made available to the Division upon request.
- The baghouse operation and maintenance requirements in Condition 25.3.1 have been removed. This condition refers to the baghouse operation and maintenance requirements in Condition 20.
- Clarified the requirements in the summary table under Condition 25.4 (Reg 1 opacity requirements).
- The opacity requirement in Condition 25.5 was revised to include the NSPS Subpart F opacity requirement.

- Added a “new” Condition 25.6 for the NESHAP LLL requirements. Note that while these sources are subject to the more stringent opacity requirement in NSPS F, they are still subject to the operation and maintenance plan, as well as any related recordkeeping and reporting requirements in NESHAP LLL.

“New” Section II.26 – Stationary Internal Combustion Engines

In October 2011, the AQCC adopted revisions to Colorado Regulation No. 3 to address APEN exemptions that EPA would not approve as SIP revisions and as a result the APEN exemptions for stationary internal combustion engines and emergency generators were removed. Such equipment would still be exempt from the APEN reporting requirements provided actual, uncontrolled emissions exceeded the de minimis level.

CEMEX submitted APENs on July 1, 2013 for several stationary internal combustion engines, presumably because they believed APENs were required. One of these engines, the A-pit pump, did have actual, uncontrolled emissions above the APEN de minimis level (1 tpy of NO_x) and thus an APEN was required. APENs were not required for the other four engines, because actual, uncontrolled emissions were below the APEN de minimis level. The APENs for the four engines were not logged in or processed since APENs were not required.

Stationary internal combustion engines that have actual, uncontrolled emissions below 5 tons/yr can still be considered an insignificant activity, even if an APEN would be required (see Colorado Regulation No. 3, Part C, Section II.E.3.nnn.(iii)).

As discussed previously in this document, these engines are also subject to requirements in the RICE MACT and under the “catch-all” provisions in Colorado Regulation No. 3, Part C, Section II.E, sources that are subject to any federal or state applicable requirement, such as MACT requirements cannot be considered insignificant activities. As a result all of the engines included in the July 1, 2013 submittals need to be included in “new” Section II.26 of the permit.

It should be noted that the source indicated that the emergency fire pump engine, which was included in the July 1, 2013 APEN submittal is no longer operable, thus that that engine will not be included in the permit.

The source did note that there is an emergency engine that supplies power to the flood response pumps. This engine is subject to the RICE MACT requirements and also the requirements in 40 CFR Part 60 Subpart IIII. Information on this engine was provided in an October 18, 2016 email. The description for the new engine is as follows:

Cummins Model No. 4BTAA3.3G7, S/N 75021552, diesel fuel-fired engine providing emergency power to flood response pumps. The engine is rated at 99 hp (73.8 kw), displacement 0.825 liters/cylinder (total displacement 3.3 liters) and 4.2 gal/hr. The engine was manufactured on May 25, 2015.

The descriptions for the engines addressed in the July 1, 2013 APEN submittals are as follows:

John Deere, Model No. 4.5L, S/N unknown, diesel fuel-fired engine driving a water pump (A-pit pump). This engine is rated at 90 hp and 0.7 MMBtu/hr (4.7 gal/hr)

John Deere, Model No. T0404045DF150, S/N unknown, diesel fuel-fired engine driving a water pump (Dowe Flats 6 in pump). This engine is rated at 80 hp and 0.6 MMBtu/hr (4.2 gal/hr).

John Deere, Model No. T0404045DF150, S/N T04045T532755, diesel fuel-fired engine driving a water pump (Dowe Flate 8 in pump). This engine is rated at 84 hp and 0.6 MMBtu/hr (4.4 gal/hr).

Natural gas-fired engine used to provide kiln rotation during power failure. No make, model or serial no. available for this engine. This engine is rated at ~ 230 hp.

The appropriate applicable requirements for these engines are as follows:

- Except as provided for below, visible emissions shall not exceed 20% opacity (Reg 1, Section II.A.1)
- Visible emissions shall not exceed 30% opacity, for a period or periods aggregating more than six (6) minutes in any sixty (60) minute period, during fire building, cleaning of fire boxes, soot blowing, start-up, process modifications, or adjustment or occasional cleaning of control equipment (Reg 1, Section II.A.4)

Based on engineering judgment, the Division believes that the operational activities of fire building, cleaning of fire boxes and soot blowing do not apply to diesel engines. In addition, since this engine is not equipped with control equipment the operational activities of adjustment or occasional cleaning of control equipment do not apply to this engine. Finally, based on engineering judgment, it is unlikely that process modifications will occur with this engine. Therefore, for this unit the 30% opacity provision only applies during startup. The 20% opacity requirement (noted in the above bullet) applies at all other times. Note 40 CFR Part 63 Subpart ZZZZ § 63.6625(h) specifies that startup shall not exceed 30 minutes.

- SO₂ emission shall not exceed 0.8 lbs/mmBtu (Reg 1, Section VI.B.4.b.(i)).

Note that for the emergency flood response pumps, the SO₂ limit has been streamlined (included in Section III.3 of the permit) since the NSPS IIII fuel requirements are more stringent (15 ppm S, which is equivalent to 1.55×10^{-3} lb/MMBtu).

- 40 CFR Part 60 Subpart IIII requirements **(for the flood response engine only)**

The emergency engine powering the flood response pumps is subject to the Subpart IIII requirements, which includes emission and fuel limitations, as well as operating requirements. Compliance with the Subpart IIII requirements can be met by purchasing a certified engine.

- 40 CFR Part 60 Subpart A, general provisions **(for the flood response engine only)**

- Circumvention (§ 60.12)

40 CFR Part 60 Subpart IIII § 60.4218 identifies the general provisions that apply. According to the table, the provisions in § 60.7 (notification and recordkeeping) apply as specified in § 60.4214(a) and this section does not apply to these engines, therefore, the provisions in § 60.7 do not apply. The table also indicates that § 60.8 (performance testing) and § 60.13 (monitoring requirements) only apply to engines with a displacement greater than or equal to 30 liters per cylinder and therefore do not apply to these engines. In addition, the table indicates that the provisions in § 60.11 do not apply as the requirements are specified in Subpart IIII.

- 40 CFR Part 63 Subpart ZZZZ requirements

The diesel fuel-fired engines addressed in the July 1, 2013 APEN submittal are existing engines < 100 hp (commenced construction before June 12, 2006), located at a major source of HAPS and are only subject to management practices (oil and filter change, inspect air cleaner and inspect hoses and belts).

Note that the natural gas fired engine is considered an emergency engine and is also only subject to management practices.

The emergency engine powering the flood response engines, meet the Subpart ZZZZ requirements by meeting the NSPS IIII requirements.

- 40 CFR Part 63 Subpart A requirements

Since these engines are not subject to any emission limitations, monitoring requirements, notification and reporting requirements the requirements in §§ 63.7, 63.8, 63.9 and 63.10 do not apply. In addition, because the engines are existing the requirements in § 63.5 (preconstruction review and notification requirements) do not apply. Finally, Table 8 of Subpart ZZZZ indicates that operation and maintenance requirements in 63.6(e) do not apply. Therefore, the permit will only include the prohibition and circumvention requirements in § 63.4.

Section III – Permit Shield

- The following requirements were included in the permit shield for streamlined conditions (Section III.3):

- The RH recordkeeping requirements in Reg 3, Part F, Section VII.D.
- The RH reporting requirements in Reg 3, Part F, Section VII.E.
- The RH PM emission limit for the kiln (0.275 lb/ton of dry feed) in Reg 3, Part F, Section VII.A.2.
- The RH kiln PM monitoring requirements in Reg 3, Part F, Section VII.C.2.a, except for the last paragraph for opacity monitoring.
- The kiln PM limit in 12BO444-2 (0.275 lb/ton of dry feed).
- The PC MACT opacity requirements for equipment other than the kiln or clinker cooler.
- The Reg 1 SO₂ requirements for the emergency flood response engine.

Appendices

- The Reg 3 citation for the Responsible Official on the certifications in Appendices B and C were revised (the version date was also changed). In addition, the condition numbers referenced in Appendix B were corrected (no version date change).
- The following changes were made to the table in Appendices B and C:
 - Added the diesel and natural gas-fired engines.
- Revised Appendix D to correct EPA address (compliance notifications) and to clarify permit mods sent to EPA.
- Cleared the table in Appendix F.

Potential to Emit

Note that values in bold brackets “[]” in the table below reflect the PTE when the new PC MACT and Regional Haze (RH) requirements take effect. Note that the PC MACT total hydrocarbon (TCH) limits (kiln and dryer) and HCl limits (kiln) are not reflected as these are ppm limits and not easily reduced to a tons/yr limit.

Process	Plant ID	AIRS Pt No	Description	Basis for PTE	Emissions (tons/yr)							
					PM	PM10	SO2	NOX	CO	VOC	Pb	HAPs ¹
Dowe Flats/Lyons Quarry - Fugitive Emissions	P017	017	Blasting - combustion by-products	permit limits			1.2	10.0	39.6			
		025	Fugitive PM - Dowe Flats	permit limits	134.2	58.4						
			Fugitive PM - Lyons Quarry, disturbed areas	permit limits	19	9.4						
Dowe Flats Quarry - Pt Source Emissions		026	Conveyor	permit limits (includes both crusher and conveyor)	0.16	0.07						
		027	Primary Crusher									
General Fugitive	P018	028	Process Fugitives - Lyons Plant	2009 actuals x 1.2	47.33	6.54						
		019	Haul Roads - Not Subject to Emission Limitations	2012 actuals x 1.2	22.96	9.3						
Raw Material Storage and Handling at Plant Site	P000	024	Disch of primary crushed mtl/front end loader	permit limits	15.5	7						
Primary Crusher (plant)	P001	001	Primary crushing/surge silo	emission factors in T5 permit and throughput limits on related equip.	0.54	0.54						
Raw Mtls Drying	P002	002	Dryer	permit limits	22.8	22.8	36.7	13.9	57.3	144.8	1.6	
Secondary Crushing	P003	003	secondary crushing/screening, belt trans.	emission factors in T5 permit and throughput limits on related equip.	0.18	0.18						
Raw Mtl Storage Silos	p004	004	raw mtl silos	emission factors in T5 permit and throughput limits on related equip.	0.048	0.048						
Raw Mtl Grinding	P005	005	raw mtl grinding	emission factors in T5 permit and throughput limits on related equip.	33.15	33.15						

Process	Plant ID	AIRS Pt No	Description	Basis for PTE	Emissions (tons/yr)							
					PM	PM10	SO2	NOX	CO	VOC	Pb	HAPs ¹
Homogenizing and Blending	P006	006	silos	emission factors in T5 permit and throughput limits on related equip.	0.032	0.032						
Kiln Burning	P007	007	Kiln	permit limits and/or future regulatory requirements	133 [21]	133 [21]	1,340 [95]	2,649 [901]	396	138	4.4	11.43
Clinker Cooler	P009	008	Clinker Cooler	MACT LLL limit (1999/ [2015]) and kiln feed limits. Assume PM = PM10	48.4 [21]	48.4 [21]						
Clinker & Gypsum/Additive Silos and Weigh Feeders	P009	009	feeders, silos, transfer belts	permit limits	9.3	4.65						
Sheltered (A-Frame) Clinker Storage and Reclaim	P010	010	Reclaim Feeder, A-Frame, Transfer belts	permit limits	21.96	10.98						
Outdoor Clinker Storage & Handling	P015	015	Outdoor clinker storage pile	permit limits	2.05	0.92						
Cement Finish Mill and Auxiliaries	P011	011	Finish Mill	permit limits	17.05	8.65						
	P012	031	Finish Mill Separator	permit limits	21.9	10.95						
Cement Silos/Packhouses/ Loadout	P013	013	silos, packhouses	permit limits	12.3	6.2						
Material Handling System - Load-in and Load-out	P014	014	Silo, loader - coal and clinker	emission factors in T5 permit and throughput limits on related equip.	0.005	0.005						
Handling and Processing of CKD & Raw Mtl Waste Dust	P007A	049	dust silos	permit limits	19.95	9.98						
			pug mill/truck loading	permit limits	2.66	2.66						
			hauling and disposal	permit limits	5.5	2.5						
Rail Car Unloading	P050	050	Rail Car Unloader and Handling System	permit limits	0.6	0.4						
Lime Storage Silo	LIS-1		Lime Silo	requested limits	0.052	0.052						
Lime Weigh Hopper	LIS -2		Lime Weigh Feeder	requested limits	0.052	0.052						
				Total	590.68 [451.28]	386.86 [247.46]	1,377.90 [132.90]	2,672.90 [924.90]	492.90	282.80	6.00	11.43

¹HAPs from the kiln include only HCl emissions and are based on the emission factor included in the October 23, 1998 revised T5 application and the allowable clinker production limit (600,000 tons/yr)

Actual Emissions

Process	Plant ID	AIRS Pt No	Description	Data Year ¹	Emissions (tons/yr)						
					PM	PM10	SO2	NOX	CO	VOC	HAPS ²
Dowe Flats/Lyons Quarry - Fugitive Emissions	P017	017	Blasting - combustion by-products	2012			0.44	3.78	14.91		
		025	Fugitive PM	2012	97.3	31.7					
Dowe Flats Quarry - Pt Source Emissions		026	Conveyor	2009	0.03	0.01					
		027	Primary Crusher		0.092	0.041					
General Fugitive	P018	028	Process Fugitives - Lyons Plant	2009	39.44	5.45					
		019	Haul Roads - Not Subject to Emission Limitations	2012	19.13	7.75					
Raw Material Storage and Handling at Plant Site	P000	024	Disch of primary crushed mtl's/front end loader	2012	9.3	4.2					
Primary Crusher (plant)	P001	001	Primary crushing/surge silo	2012	0.34	0.34					
Raw Mtl's Drying	P002	002	Dryer	2012	7.15	3.98	1.85	3.94	17.83	3.01	0.93
Secondary Crushing	P003	003	secondary crushing/screening, belt trans.	2012	0.11	0.11					
Raw Mtl Storage Silos	p004	004	raw mtl silos	2012	0.009	0.009					
Raw Mtl Grinding	P005	005	raw mtl grinding	2012	20.62	20.62					
Homogenizing and Blending	P006	006	silos	2012	0.019	0.019					
Kiln Burning	P007	007	Kiln	2012	22.56	22.56	21	1,041.95	273.35	1.9	3.5
Clinker Cooler	P009	008	Clinker Cooler	2012	13.39	11.58					
Clinker & Gypsum/Additive Silos and Weigh Feeders	P009	009	feeders, silos, transfer belts	2012	3.61	1.8					
Sheltered (A-Frame) Clinker Storage and Reclaim	P010	010	Reclaim Feeder, A-Frame, Transfer belts	2009	9.08	4.54					
Outdoor Clinker Storage & Handling	P015	015	Outdoor clinker storage pile	2011	2.03	0.18					

Process	Plant ID	AIRS Pt No	Description	Data Year ¹	Emissions (tons/yr)						
					PM	PM10	SO2	NOX	CO	VOC	HAPS ²
Cement Finish Mill and Auxiliaries	P011	011	Finish Mill	2009	5.24	2.62					
	P012	031	Finish Mill Separator	2009	10.64	5.32					
Cement Silos/Packhouses/ Loadout	P013	013	silos, packhouses	2009	2.02	1.01					
Material Handling System - Load-in and Load-out	P014	014	Silo, loader - coal and clinker	2011	0.003	0.003					
Handling and Processing of CKD & Raw Mtl Waste Dust	P007A	049	dust silo, pug mill, loading, truck hauling & Disposal	2012	4.8	2.53					
Rail Car Unloading	P050	050	Rail Car Unloader and Handling System	2010	0	0					
Lime Storage Silo	LIS-1		Lime Silo	PTE	0.052	0.052					
Lime Weigh Hopper	LIS -2		Lime Weigh Feeder	PTE							
				Total	266.97	126.42	23.29	1,049.67	306.09	4.91	4.43

¹For 2009 data year, based on APENs rec'd 4/29/10; for 2010, based on APENs rec'd 11/14/11; for 2011, based on APENs rec'd 4/30/12; for 2012 data year, based on APENs rec'd 4/29/13; and for PTE, based on requested emissions.

²Reported HAPs from the dryer are Pb. HAPs reported that exceed the APEN de minimis level for the kiln are HCl, acetaldehyde and carbon disulfide.

CAM Applicability Analysis

The CAM requirements apply to any emission unit that uses a control device to meet an emission limitation or standard and has pre-controlled emissions above the major source level. The following addresses the CAM applicability, with respect to particulate matter (PM/PM₁₀) emissions, of the various emission sources or activities addressed in this permit.

Emission sources or activities with no emission limitations

In general all of the emission sources or activities addressed in the permit except for those listed in the table below have some sort of emission limitation. Since these sources or activities do not have emission limitations, CAM does not apply to these emission units.

Process (Permit Section)	Plant ID	AIRS ID	Description	Title V Permit Section
General Fugitive Emissions Requirements	P018	028	Process Fugitives (Lyons Cement Plant) Not Subject to Emission Limitations Includes wind erosion of stock piles and various transfers not vented through a stack (e.g. belt and screw conveyor transfers)	Section II.14
		019	Haul Roads (Lyons Cement Plant/Quarry and Dowe Flats Quarry) Not Subject to Emission Limitations Hauling of purchased limestone, iron, gypsum and silica and operation of water truck	
Material Handling System – Load-In & Load-Out	P014	014	Outdoor Coal Storage	Section II.12
Cold Cleaner Solvent Vats		APEN Exempt	Cold Cleaner Solvent Vats	Section II.18
Gasoline Storage Tank		APEN Exempt	Gasoline Storage Tank (3,000 gallons, aboveground)	Section II.15
Stationary Internal Combustion Engines	A-Pit Pump	053	John Deere, Model No. 4.5L, diesel fuel- fired engine driving a water pump. This engine is rated at 90 hp and 0.7 MMBtu/hr (4.7 gal/hr)	Section II.27

The remaining emission units are subject to emission limitations and thus it is necessary to determine if CAM applies.

Emission sources with emission limitations

No control device

A number of emission units or activities rely on work practices or control measures (e.g. watering, application of chemical stabilizers, limited speeds) to reduce emissions. However, the Division does not consider that these types of control measures meet the

definition of control equipment. The preamble to the CAM rule provides more insight into the control technology definition and provides the following (from October 22, 1997 Federal Register, page 54912, 3rd column, under *control devices criterion*)

The final rule provides a definition of “control device” that reflects the focus of Part 64 on those types of control devices that are usually considered as “add-on” controls.” This definition does not encompass all conceivable control approaches but rather those types of control devices that may be prone to upset and malfunction, and that are most likely to benefit from monitoring of critical parameters to assure that they continue to function properly. In addition, a regulatory obligation to monitor control devices is appropriate because these devices generally are not a part of the source’s process and may not be watched as closely as devices that have a direct bearing on the efficiency or productivity of the source.

The Division considers that the use of work practices or control measures (e.g. watering, application of chemical stabilizers, limited speeds) to reduce fugitive and/or visible emissions are not considered add-on control devices and are not the type of devices that would benefit from monitoring critical parameters. Therefore, the Division considers that CAM does not apply to the sources listed in the table below as these sources or activities do not use a control device.

Process (Permit Section)	Plant ID	AIRS ID	Description		Title V Permit Section
Dowe Flats and Lyons Quarry – Fugitive Emission Sources	P017	017	Blasting (combustion byproduct emissions)		Section II.1
		025	(Particulate Emissions Only) Drilling , Blasting, Truck Loading/Unloading, Haul Roads (Dowe Flats), Scraper Activities, Grading, Bulldozing, Wind erosion of stockpiles and exposed areas		
Raw Material Storage and Handling at Plant Site)	P000	024	Discharge of Primary-Crushed Material onto Open Stockpile S009 - Front End Loader Activity		Section II.3
Outdoor Clinker Storage and Handling	P015	015	Outdoor Clinker Storage Pile		Section II.11
Handling and Processing of CKD and Raw Material Waste Dust	P007A	049	041 - Pug Mill Mixing, Pelletization and Truck Loading of CKD and Benefication Dust	041 - Pug Mill/Truck Loading	Section II.13
			042 - Haulage and Disposal of Pelletized CKD and Benefication Dust	042 - Truck Hauling and Disposal at Lyons Quarry	

Sources with control devices

Emission sources with no annual emission limitations

Clinker cooling and transfer to storage for finish mill

Process (Permit Section)	Plant ID	AIRS ID	Description	Title V Permit Section
Clinker Cooling and Transfer to Storage for Finish Mill	P008	008	S017 – Clinker Drag Chains (1 baghouse)	Section II.10
			S018 - Clinker Cooler (2 baghouses, 1 stack)	
			S023 – 529-25 Drag Conveyor (1 baghouse)	
			S024B – Outside Clinker Drop Hood (1 baghouse, vented to S018 stack through 525-8/9)	

None of the emission points in the clinker cooler group are subject to annual emission limitations but all are subject to short term emission limits. The clinker cooler (S018) is subject to a PM emission limitation in the PC MACT and also to the process weight rate PM limits in Reg 1. Since the PC MACT PM limit is more stringent than the Reg 1 limit, the Reg 1 limit was streamlined. Emission limitations under a NESHAP proposed after November 15, 1990 are exempt from CAM (see 40 CFR Part 64 § 64.2(b)(i)), so CAM does not apply to the clinker cooler.

Uncontrolled emissions from the remaining three units (S017, S023 & S024B) under this emission group were estimated based on the emission factor in AP-42, Section 11.6 (dated January 1995), for finish grinding mill feed belt with fabric filter (0.0024 lb/ton) and the clinker production limit of 600,000 tons/yr (this production limit applies to other equipment at this facility and also essentially limits operation of this equipment). Since this emission factor includes the control device, a control efficiency of 99% has been assumed. Although baghouse control efficiencies up to 99.9% are possible, the Division is assuming 99% for baghouses at this facility because many of the baghouses at this facility are older and may not be as efficient. The Division considers that a 99% control efficiency represents a reasonable level of control for these units. Based on this calculation method, uncontrolled emissions from P017, P023 and S024B are 72 tons/yr, each. Therefore, S017, S023 and S024B are not subject to CAM since potential pre-control emissions are below the major source level.

Other sources

The sources listed below are not subject to annual emission limitations but all are subject to the process weight rate limits in Reg 1. Potential uncontrolled emissions were estimated used the emission factors listed in the Title V permit and annual production limits. Note that the equipment in question has no annual production limits but production limits for other equipment serve to limit operation of the equipment in question. When the emission factors in the permit include controls, a 99% control efficiency is presumed. The results of this analysis are provided in table 1 on page 67.

Table 1 on page 67 indicates that potential pre-control emissions from all sources except the raw mill feeders (S012), the raw mill (S010) and the raw mill separator (S011) are below the major source threshold. Therefore, those emission units are not subject to CAM.

For the raw mill feeders (S012), the raw mill (S010) and the raw mill separator (S011) in order to determine whether CAM applies it is necessary to determine if the control is necessary to comply with the Reg 1 process weight rate limits. These limits are short term (lb/hr), fairly conservative and are based on the hourly processing rate of the equipment. The first renewal permit (issued March 1, 2008) included some hourly process rates (e.g. 180 tons/hr of lime and shale and 150 tons/hr of raw meal feed) for this equipment, which results in hourly emission rates of 39.7 and 38.6 lb/hr respectively. Uncontrolled hourly emissions from the raw mill feeders (S012), the raw mill (S010) and the raw mill separator (S011) were estimated using the emission factor in the permit, a control efficiency of 99% and the allowable hourly throughput rate. As shown in the table below uncontrolled hourly emissions exceed the process weight rate limits thus CAM applies (the baghouse is necessary to comply with the process weight rate limit).

Emission Source	Emission Factor ¹	Hourly Throughput ²	Uncontrolled Hourly Emissions ³
S012 - Raw Mill Feeders	0.019 lb/ton (c)	130.2 tons	247.4
S010 - Raw Material Grinding	0.012 lb/ton (c)	130.2 tons	156.2
S011 –Raw Material Separator	0.032 lb/ton (c)	130.2 tons	416.6

¹ Emission factors with a (c) indicate that the factor is for controlled emissions. The controlled emission factors are from AP-42, Section 11.6 (dated January 1, 1995).

²The throughput limits included are limits for other emission units but limit operation of the equipment addressed in the table. The hourly limit is based on the annual limit for raw materials (1,050,000 tons/yr) divided by hours of operation for the kiln (8,064 hrs/yr), which is presumed to limit operation of the facility.

³Since the emission factors include controls, a control efficiency of 99% was presumed for the baghouses to estimate pre-control emissions.

Emission sources with annual limitations

The remaining sources have annual emission limitations for PM and PM₁₀, and in some cases daily PM₁₀ limits. Annual emissions for PM and PM₁₀ for all emission sources except the kiln are below the major source level. Therefore, the kiln is subject to CAM for PM and PM₁₀ and potential pre-controlled emissions from the other sources need to be assessed to determine CAM applicability.

Several sources rely on emission factors and throughput or hours of operation to assess emissions. Potential pre-control emissions were calculated for these sources as shown in Table 2 on page 68. For these sources, only the raw material dryer (S005) has emissions above the major source level and as such is subject to CAM.

The majority of the sources included in Section II.11 and II.13 of the permit, rely on grain-loading specifications, the baghouse design rate (in cubic feet per minute) and allowable hours of operation.

As discussed previously in this document, in the current Title V permit (last revised April 1, 2013), the grain-loading “limits” for many of the units do not ensure compliance with the emission limitations, therefore the grain-loading limits have been reduced. The pre-control emissions have been estimated based on the new, lower grain-loading “limits.”

The analysis for emission group P013 (in Section II.11 of the permit) included the removal of the south (S049) and north (S050) packers. In their September 26, 2016 comments on the draft permit the source noted that these emission units were inoperable and requested that they be removed from the permit.

The results of this analysis are shown in Table 3 on pages 69 and 70. Based on this analysis, only the following sources have emissions above the major source level:

- PM only: S024 (#2 clinker silo), S046 (packhouses west and east – 2 baghouses) and S066 (cement silo A5)
- PM and PM₁₀: S034 (#6 reclaim feeder and A-frame), S051 (Top of A frame), S036 (finish mill), S037 (finish mill auxiliary dust collector), S065 (finish mill separator – 2 baghouses), S043 (masonry silos A10 and A13), S044 (cement silo A7), S045 (cement silo A2), S001 (waste dust silo), and S022 (kiln return dust silo).

All of the above mentioned sources with pre-control PM and/or PM₁₀ emissions above the major source level, except the finish mill separator are subject to CAM. The underlying construction permit (98BO0259) for the finish mill separator indicated that the finish mill separator baghouses were inherent process equipment, therefore the finish mill separator is not subject to CAM.

Table 1 – Other Sources Not Subject to Annual Emission Limits

Process	Plant ID	AIRS ID	Description	Pollution Control Device	Title V Permit Section	PM Emission Factor ¹	Annual Throughput Limit ²	Potential Pre-Control PM Emissions ³
Primary Crusher (Plant)	P001	001	S002 - Primary Crushing (Plant)	Baghouses	Section II.4	0.001 lb/ton (c)	1,050,000 tons	52.5 tons
			S004 - Surge Silo			2.9×10^{-5} lb/ton (c)	1,050,000 tons	1.5 tons
Secondary Crushing	P003	003	Secondary Crushing and Screening (vents to S001 – Waste Dust Silo)	Baghouse (2 total)	Section II.6	0.00031 lb/ton (c)	1,050,000 tons	16.3 tons
			S003 - #4 & #6 Belt Transfer			2.9×10^{-5} lb/ton (c)	1,050,000 tons	1.5 tons
Raw Material Storage Silos	P004	004	S006 to S008 - Raw Material Storage Silos	Baghouse (3 total)	Section II.7	2.9×10^{-5} lb/ton (c) (for each stack)	1,050,000 tons	1.5 tons (for each stack)
Raw Material Grinding	P005	005	S012 - Raw Mill Feeders	Baghouse (4 total)	Section II.8	0.019 lb/ton (c)	1,050,000 tons	997.5 tons
			S013 - Iron/Silica Silo			0.0031 lb/ton (c)	525,000 tons	81.4 tons
			S010 - Raw Material Grinding			0.012 lb/ton (c)	1,050,000 tons	630 tons
			S011 –Raw Material Separator			0.032 lb/ton (c)	1,050,000 tons	1,680 tons
Homogenizing & Blending	P006	006	S014 - Homogenizing Silo	Baghouse (2 total)	Section II.9	2.9×10^{-5} lb/ton (c)	967,680 tons	1.4 tons
			S015 - Kiln Feed Silo			2.9×10^{-5} lb/ton (c)	967,680 tons	1.4 tons
Material Handling System – Load-In & Load-Out	P014	014	S020 - Coal Silo/Elevator		Section II.12	Coal or clinker: 2.9×10^{-5} lb/ton (c) Other materials: 0.0069 lb/ton	Coal or clinker: 113,945 tons Other materials: 1,050,000 tons	Coal or clinker: 0.16 tons Other materials: 3.6 tons
			S019 – Material Unloading Hopper (Railcar)			Coal or clinker: 2.9×10^{-5} lb/ton (c) Other materials: 0.0069 lb/ton	Coal or clinker: 113,945 tons Other materials: 1,050,000 tons	Coal or clinker: 0.16 tons Other materials: 3.6 tons
			S025 – Material Unloading Hopper and Spout (Trucks)	Baghouse		Coal or clinker: 2.9×10^{-5} lb/ton (c) Other materials: 0.0069 lb/ton	Coal or clinker: 113,945 tons Other materials: 1,050,000 tons	Coal or clinker: 0.16 tons Other materials: 3.6 tons

¹ Emission factors with a (c) indicate that the factor is for controlled emissions. The controlled emission factors are from AP-42, Section 11.6 (dated January 1, 1995). The uncontrolled emission factor is from AP-42, Section 11.2 (dated June 2006), for aggregate transfer.

²The throughput limits included are limits for other emission units but limit operation of the equipment addressed in the table. The limit of 1,050,000 tons/yr is for raw materials (see Sections II.2 II.3 and II.5), the 967,680 tons/yr is the limit for kiln feed (see Section II.10) and the limit of 113,945 tons/yr is for coal consumption in the kiln. Iron and silica are components in the feed and not necessarily the primary component. Therefore, since the iron/silica silo (S013) contains only a portion of the feed/raw material, it does not seem appropriate to use the 1,050,000 ton/year limit for this unit. For S013, emissions are based on 50% of the raw material limit (525,000 tons/yr).

³Since the emission factors include controls, a control efficiency of 99% was presumed for the baghouses to estimate pre-control emissions.

Table 2: - Sources with Annual Emission Limits Relying on Emission Factors or Other Information

Process	Plant ID	AIRS ID	Description	Title V Permit Section	PM/PM ₁₀ Emission Factor ¹	Annual Throughput Limit	Permitted PM/PM ₁₀ Emissions	Potential Pre-Control Emissions ²
Dowe Flats Quarry – Point Source Emissions	P017	026	S056 through S064 – Conveyor	Section II.2	PM: 0.00124 lb/ton PM ₁₀ : 0.00059 lb/ton	1,050,000 tons		PM: 0.65 ton PM ₁₀ : 0.31 tons
		027	S055 - Primary Crusher (Quarry)		PM: 0.020 lb/ton PM ₁₀ : 0.009 lb/ton	1,050,000 tons		PM: 10.5 tons PM ₁₀ : 4.7 tons
Raw Materials Drying	P002	002	S005 - Raw Materials Dryer	Section II.5			22.8 tons	2,280 tons
Cement Rail Car Unloading System	P050	050	Cement Rail Car Unloading and Handling System – Hopper, screw conveyor and pneumatic transfer system	Section II.25	PM: 1.32 lb/ton PM ₁₀ : 0.86 lb/ton	50,000 tons		PM: 33 tons PM ₁₀ : 21.5 tons
Kiln Control Device Support Equipment	LIS-1	055	BCSA Inc, Silotop R03, Lime Storage Silo, S/N unknown.	Section II.26	0.61 lb/ton	12,096 tons		3.7 tons
	LIS-2	054	BCSA Inc, Silotop R03, Lime Weigh Hopper, S/N unknown		0.61 lb/ton	12,096 tons		3.7 tons
	ACI-1	N/A	Activated carbon storage silo	Insig. Activity List (App. A)	0.61 lb/ton	202 tons		0.06 tons

¹ Emission factors with a (c) indicate that the factor is for controlled emissions. Unless otherwise specified, the emission factor source is listed in the T5 permit.

² When no emission factor is listed, potential pre-control emissions are based on permitted emissions and an assumed baghouse control efficiency of 99%.

Table 3: - Sources with Annual Emission Limits Estimated using Grain-Loading Emission Factors

Process	Plant ID	AIRS ID	Description	Title V Permit Section	Baghouse Id Number	Grain Loading (gr/dscf) ¹		Baghouse Design (dscfm) ²	Potential Pre-Control Emissions (tons) ³	
						PM	PM ₁₀		PM	PM ₁₀
Clinker and Gypsum/ Additive Silos and Weigh Feeders (Storage and Transfer to Finish Mill)	P009	009	S021 – Top of A Frame (Belt 529-30 to 529-63)*	Section II.11	525-15	0.0233	0.0117	585	47.2	23.6
			S026, S027, S029, S030, S031 – Weigh Feeders 1, 2, 4, 5 and 6*		625-4	0.0233	0.0117	605	48.8	24.4
					625-5	0.0233	0.0117	605	48.8	24.4
					625-7	0.0233	0.0117	605	48.8	24.4
					625-8	0.0233	0.0117	605	48.8	24.4
					625-9	0.0233	0.0117	605	48.8	24.4
			S024 - #2 Clinker Silo		625-12	0.0233	0.0117	1,678	135.3	67.7
			S032 – Bottom of A Frame Transfer*		625-10	0.0233	0.0117	1,210	97.6	48.8
			S033 Gypsum/Limestone from 529-31 belt to Silos		625-11	0.0233	0.0117	839	67.7	33.8
			S035 – Discharge of 629-3 Belt		625-14	0.0233	0.0117	839	67.7	33.8
			S039 to S041 –Finish Mill Weigh Feeders**		725-6	0.0233	0.0117	839	67.7	33.8
					725-7	0.0233	0.0117	839	67.7	33.8
					725-8	0.0233	0.0117	839	67.7	33.8
			S038 - Surge Bin**		725-5	0.0233	0.0117	839	67.7	33.8
			*stacks vent inside A-Frame							
*stacks vent inside mill building.										
Sheltered (A-Frame) Clinker Storage and Reclaim	P010	010	S034 - #6 Reclaim Feeder and A-Frame Building	Section II.11	625-15	0.0146	0.0073	37,538	945.6	1,891.2
			S051 – Top of A Frame – Transfer from 529-29 belt to 529-30 belt		525-17	0.0146	0.0073	6,049	152.4	304.8
Cement Finish Mill and Auxiliaries	P011	011	S036 - Finish Mill	Section II.11	725-2	0.0215	0.0102	11,335	841.8	398.1
			S037 – Finish Mill Auxiliary Dust Collector		725-3	0.0215	0.0102	11,623	863.2	408.3

Process	Plant ID	AIRS ID	Description	Title V Permit Section	Baghouse Id Number	Grain Loading (gr/dscf) ¹		Baghouse Design (dscfm) ²	Potential Pre-Control Emissions (tons) ³	
						PM	PM ₁₀		PM	PM ₁₀
Cement Finish Mill and Auxiliaries	P012	031	S065 – Finish Mill Separator	Section II.11	725-10	0.0058	0.0029	53,507	1,075.9	538
			S065 – Finish Mill Separator		725-11	0.0058	0.0029	53,507	1,075.9	538
			S069 - Clinker Dust to Finish Mill (SEP project) – vents inside mill room		725-28	0.01	0.005	1,104	38.2	19.1
Cement Silos/ Packhouse/ Loadout	P013	013	S043 – Cement Storage Silos A10 and A13	Section II.11	825-1	0.0239	0.0120	3,438	283.9	143.1
			S044 – Cement Storage Silo A7		825-2	0.0239	0.0120	3,438	283.9	143.1
			S045 – Cement Finish Silo A2		825-3	0.0239	0.0120	3,438	283.9	143.1
			S046 – Packhouses West and East (Loading Spouts, Baghouses 825-4 and 825-5 vent to a common stack)		825-4	0.0239	0.0120	1,376	113.6	57.3
			S046 – Packhouses West and East		825-5	0.0239	0.0120	2,131	176.0	88.7
			S048 – Recirculating System		825-6	0.0239	0.0120	1,074	88.7	44.7
Handling and Processing of CKD and Raw Material Waste Dust	P007A	049	S066 Cement Silo A5	Section II.13	525-28		005	2,985	103.2	51.6
			S067 - CKD Loading Spout (vents indoors)		825-7		005	2,181	75.4	37.7
			S001 - Waste Dust Silo		225-3		0.005	12,968	448.2	224.1
			S022 - Kiln Return Dust Silo		525-21		0.015	3,219	333.8	166.9

¹The grain-loading “limits” or “emission factors” shown in this table were revised and are included in the draft permit and were calculated to ensure compliance with permitted emissions.

²The baghouse design rates indicated in the Title V permit are in “actual cubic feet per minute”. The baghouse design rates in “dry standard cubic feet per minute” were taken from the information provided by CEMEX for the Division’s 2014 annual inspection. The stack temperature and pressures are not expected to change significantly, thus relying on this information is acceptable.

³ Potential pre-control emissions are based on emissions calculated from the grain loading specification, the baghouse design rate, allowable hours of operation (8,064 hrs/yr) and an assumed baghouse control efficiency of 99%.